

Mus 320 The Life & Works of Bach

Numbered Transcript

MUS 320 • Samuel Andreyev

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Lecture 1: Bach's Musical World

At a glance — Andreyev opens his course on J.S. Bach by asking 'Why Bach?' and sketching Bach's musical world — his family of musicians, the Lutheran context, and the biographical arc from Eisenach through Weimar, Köthen, and Leipzig that frames the works to come.

- 1.1 Why Bach? Of all the composers that I could have chosen to devote an entire course to, he was the obvious choice. And this is something that I've been reflecting on for the past few days. Why was he the obvious choice? There are no shortage of remarkable figures in the history of Western art music. We know their names. Mozart comes to mind. Beethoven, Haydn, Brahms, Schumann, Schubert, Schönberg, Stravinsky. It's a distinguished lineage. And somehow, out of all of those figures, Bach was the obvious choice. It didn't even require much reflection.
- 1.2 So what is it about Bach that puts him at the center of my musical preoccupations and those of so many other musicians? It's not to say that he's the greatest of all, that his achievement cannot be surpassed, although some would make that argument. It's more like he lies at the center. And I think the reason for that is, well, there's a few things. The first is that the breadth of his achievement is simply staggering. So much so that it's almost impossible for us to comprehend how he could have done what he did.
- 1.3 We've all heard stories of the extraordinary facility and fluidity of Mozart's pen. You know, how he could apparently dream up a piece in his head and then writing it down was just a perfunctory matter—you know, it was all there. We've all heard amazing stories about the seemingly unstoppable productivity of Haydn. Vivaldi, with his hundreds of concertos, five or six hundred of them altogether. Scarlatti with his hundreds of sonatas. Telemann, who is a contemporary of Bach's, wrote about 5,000 pieces in his lifetime.
- 1.4 But there's really something unique about the achievements of Johann Sebastian Bach. And I've been puzzling over this for basically my entire life as a musician. When I say the "breadth of his achievement," it's not just a question of how many pieces he was able to complete. It's something to do with the universality of his music. The fact that it's both the most abstract music I can think of, in many respects, and also the most immediately expressive. His capacity to combine those two things in one and the same work is astounding.
- 1.5 Many musicians have remarked that you can find everything in Bach. Beethoven by the time he was 11 years old had memorized The Well-Tempered Clavier of Bach, which is a gigantic compendium of pieces, a kind of encyclopedia of everything you can do in fugue, and Beethoven said of Bach, "He is the Urvater of harmony," the forefather or the father of all harmony. Amazing.
- 1.6 The Austrian composer Anton von Webern—who is known for combining a fascination of early music with the Franco-Flemish polyphonic style, with a very modernist

tendency to compress and condense, an intense expressionist language—said, "You can find everything in Bach. Nothing's missing." And in my experience, it's very difficult to think of anything that you're left wanting for. When you engage with his work, it's an amazing thing.

- 1.7 So it's something like the breadth and the depth of his achievement, the profundity of the work, the fact that it appears to be inexhaustible. What does it mean for a work of art to be inexhaustible? Well, to me, it means you can engage with it your entire life and keep finding new things in it. And not only that, it speaks to us generation after generation after generation. You can't exhaust it.
- 1.8 There's a kind of a trope in the music world, in the art world generally, which is the idea of the artwork that hasn't aged well, and we all know what that means. There are works that, for one reason or another, maybe they're not necessarily intended to last very long. Maybe they're so journalistic and ephemeral in their preoccupations that once their moment has passed, there's no particular reason to bother with them anymore.
- 1.9 And indeed, the libraries, the great libraries and archives of Europe, are filled to the brim with countless manuscripts by Baroque and Classical composers that there's no particular reason to listen to again. Maybe as a curiosity every now and then, but there are many pieces that are essentially interchangeable. In other words, they contain a minimal degree of individuality, a minimal degree of true invention.
- 1.10 You have the impression with Bach that even the most minor work, even works that were intended to have a didactic function—you know, pieces for kids to play, for example, or pieces to demonstrate a technical principle—even works like that have this extraordinary expressivity, this depth to them.
- 1.11 I'll tell you a story about how I first came to Bach and what the music meant to me. So growing up, I was fascinated by all aspects of the musical phenomenon instinctively. And I was always looking for things to listen to. And, you know, as you do when you're a kid, you just grab anything and you try it and you see. So I listened to whatever pop music was on the radio. I listened to my sister's record collection—they grew up in the 80s, so they had different tastes than me, but I listened. My parents had some classical music LPs, and I listened to those too.
- 1.12 And good Canadian that I was I, of course, was interested in Glenn Gould, who's one of our national treasures, one of our great artists. And if you're interested in music, especially classical music, if you were growing up in Toronto, it's hard not to take an interest in Glenn Gould. And so I got a hold of some Glenn Gould recordings, and I'd heard that he was a great Bach specialist. So I started with that and I got recordings of the Two- and Three-Part Inventions (Sinfonias), the Italian Concerto, The Well-Tempered Clavier, Art of Fugue, all of these great pieces, and started listening.

- 1.13** I was especially taken with the Two- and Three-Part Inventions when I started listening to his recordings of them— we're going to talk about those pieces not in this lecture, but a little bit later today. The thing that is remarkable to me about them is that these are didactic works, as I mentioned, which is to say that they were explicitly formulated around a desire to explain a compositional principle and to be useful to beginners. So that doesn't sound extraordinarily exciting when you hear about it. It's like, okay, "Here are some pieces that will help you develop finger independence. And we'll sort of lay out how to make a well-organized piece of music with two voices."
- 1.14** And in the hands of any other composer of that period, the pieces would certainly have been competent, but not necessarily particularly profound or interesting. But listening to Glenn Gould play them—remember, Glenn Gould was one of the great virtuosos of his time, he could play anything, he had technique to burn. But he worked on these little pieces and played them so divinely that it was transcendent. I was so fascinated by these pieces with their clarity of line, the clarity of design, the fact that the slightest little idea in them was striking and memorable and expressive and exciting.
- 1.15** And then I wanted to try to play them myself. And as you're going to hear, I'm going to play a few things at the piano. I'm not a pianist, I'm not a professional pianist. I play for myself. And I play also as a way of having a connection with this music. And every day when I get to my studio in the morning, the first thing I do is I play Bach at the piano. And I'm not alone in that. A lot of my fellow composers do this, too. It's a kind of a ritual.
- 1.16** So I got hold of the scores, I started working on the Inventions and Sinfonias and some of the Goldberg Variations—I can't play all of them, but some of them are within my grasp as a pianist—and The Art of Fugue. What I discovered was: when you actually sit down and play these pieces, when you have to have them in your muscle memory, when you turn them into not just an object of aesthetic contemplation or something abstract, but something embodied, when it turns into a gesture, when you're thinking in terms of the phrase the breath, you enter into a private communion with the work of this man who died in 1750.
- 1.17** And it's that one-to-one relationship with this great artist that is maybe the most undefinable thing about his work. And it has accompanied me for my entire life so far. And there's no reason to imagine that this won't continue. In fact, if anything, the deeper I get into his music, the more I realize that it's just the tip of the iceberg. It just keeps going and going. There are always more layers to it. There's always more to discover in these pieces, there's always more to learn.
- 1.18** And not only that, you have to contend with the sheer volume of his output. And every time I hear a new piece for the first time, I'm amazed by how fresh it is, how profound, how he's inventing something every time he makes a piece. And that capacity for seemingly endless expression, invention, and also pushing the possibilities of his art to the absolute limits is a source of continuous amazement.

- 1.19** Bach has a bit of a reputation as a somewhat cerebral or dry composer, maybe because a lot of the pieces that are played the most often are things like The Art of Fugue or The Well-Tempered Clavier. We hear terms like fugue and canon, and we think, "Oh, this sounds very technical." It's not something that you can immediately hang a story on.
- 1.20** When you listen to pieces by Beethoven and when you hear about what his life was like and what the music is like, you can associate that with a personality, with this sort of titanic struggle that he embodies in his work. This extreme expressiveness, this vehemence. There's something very human about it. It's much harder to humanize Bach in that sense because he's very remote from us. At least it seems that way.
- 1.21** So in this course, what we're going to do is I'm going to take you through a brief sketch of his life. There's not that much to say, actually, partly because the picture we have of Bach the man is sketchy, to say the least. We don't know very much. Bach himself was notoriously tight-lipped about his own life, didn't particularly have much interest in talking about it. Only a handful of letters have survived. So most of what we know about him comes from second-hand sources. And on top of that, it would appear that his life was not particularly eventful. So I'm going to nevertheless try to find some clues as to what sort of person he was—what would have driven him to compose the way he did.
- 1.22** But many scholars and authors and performers have observed that the best way to get to know Bach is through the music. The biographical details are not necessarily all that helpful. What we're going to do is I'm going to take you through some pieces that not only have an intense personal resonance for me, but that also illustrate different facets of Bach's work, and that also illustrate different periods of his life.
- 1.23** So now we come to a point that separates Bach out somewhat from the composers of the later Classical and Romantic eras. In Bach's time, if you were a professional musician, composer, working within what is now Germany, and was at that time an unbelievably complicated jigsaw puzzle of little duchies and principalities, you would have had to compose your work, for the most part, in accordance with what was asked of you professionally by either the court, or a cathedral, or some form of municipal employment. So you weren't necessarily free in what you got to write.
- 1.24** As Bach advanced through his career and took different posts in different cities, he had different obligations and different professional possibilities. And that conditioned to a large extent what he actually wrote. It was only really in the last decade of his life that he had the freedom to pursue projects that were a little bit more—what would you say?—that freed him from the obligation to produce constantly.
- 1.25** So let's talk a little bit about Bach, who he was, what he came from, and what his milieu was like. Bach was born in 1685 in pretty much the exact geographic center of what is now modern-day Germany. He was born in Eisenach, which is a relatively small town in Thuringia, into a vast musical family. This is a point that I think is really crucial to

understand. Bach didn't spring out of nowhere. The Bach family was very well known at that time for producing an exceptional number of musicians.

- 1.26 Pretty much every member of his family was a musician. They were town pipers, they were oboists, they were composers. His uncles, his father, his grandfather, his brother —there were seemingly dozens of members of the Bach family, all of whom had some degree of involvement with music as a profession. So you could say that the ground was fertile, in a sense. The development of the Bach family of musicians continued. It continued with him, it continued with his children, and it continued with even the extended members of the Bach family: his cousins, his second cousins, his uncles. So he didn't arise out of nowhere.
- 1.27 But his life was also full of death, full of tragedy in a way that was not uncommon at that time, but that nevertheless conditioned him to a large extent. He was an orphan by the time he was 10 years old, and his brother took over his musical education and was really his first teacher.
- 1.28 Even as a child, Bach exhibited this uncanny obsession, this profound desire to work, to learn, to improve himself in a way that was extraordinarily tenacious. There's a couple of anecdotes that have come down that illustrate this somewhat. So he wanted to copy out an organ score that his brother had kept under lock and key in an armoire in their house. So he found a way to get his hand through the grill work of the armoire, managed to roll up the score and extract it out, and then copied it out by candlelight. He was reprimanded by his brother when he'd found out that Bach had done this. But it was—nothing would stop him, and that got him into trouble on many occasions.
- 1.29 He had this aspect to his personality, which was something like stopping at nothing to go as far as possible with whatever he was doing. So he improved his skills as a musician through obsessive and relentless application. And by the time he was 15, he was already an excellent organist, an excellent improviser and a very skilled composer.
- 1.30 So here we come again to the multifaceted nature of Bach's immense gifts. He wasn't necessarily famous during his lifetime as a composer, but rather as one of the more skilled improvisers on the organ. He was justly famous for his capacity to extemporize at the keyboard. You could give him a fugue subject, a little theme, and he would instantly come up with a dizzying array of different figurations based upon that idea. He could improvise a whole piece based around it. So he was an uncommonly determined person, animated by a profound sense of professional duty, an obsession with the limits of his art, and with always going as far as possible.
- 1.31 He takes up his first post in 1703, which lasts just for a few months. This is in Weimar, which is very close to where he was born, as you'll see, there's a pattern here. Bach lived his entire life in a very small geographic region, never really leaving what is now central Germany. I mean, the exact geographical center of the country.

- 1.32** After that, he takes up a position in Arnstadt, also very close by, where he stays for a few years. In case you're thinking that he emerged spontaneously as a total consummate master of everything, that's not the case. The work does get better over the years. He does get more sophisticated, and I would say that the first real flowering of his genius comes in his early 20s. He moves to Weimar and starts to write prolifically for organ.
- 1.33** It's around this time that we start to see one of the constants of Bach's work—and this is something we're going to come to again and again—which is his capacity to take the smallest musical idea and elevate it to the highest possible degree of his art through an uncanny combinatorial mind. This is the extreme far right end of the bell curve capacity for associative thinking and the ability to instantly imagine all the possibilities of an idea, run through him, in his mind, imagine what the far-ranging consequences would be if he did this or that. That's something that very few people can do. It's extraordinarily impressive.
- 1.34** For example, when he starts writing these organ pieces, he'll take a bit of Protestant plain song. These are Lutheran hymn tunes—we're going to come back to this again and again—and build an entire edifice around these pieces, a contrapuntal elaboration. That's where we start to see the particular genius of Bach.
- 1.35** Other composers did that, too, of course, but he took it much farther. He would do things like have a hymn tune in a canonic relationship with itself, where you hear the first bar of the melody, then a bar later it starts again at a fixed interval from the first one, and when you put it together, it harmonizes perfectly. And then on top of that, he would add all kinds of figurations and contrapuntal elaborations that themselves are copies of each other and move at different speeds through the score. It's dizzying.
- 1.36** I mentioned that we don't have a lot of clues as to his personality. We don't exactly know what he was like as a father, what he was like as a husband, as an employee, but there are some clues. So he got into trouble in the most amazing way. While he was employed at Arnstadt in 1707, he decided to take a little trip to go hear the organist Dieterich Buxtehude. Buxtehude, unfortunately for Bach, lived in Lübeck, in what is now Schleswig-Holstein, in northern Germany, and Bach was in Arnstadt, over 400 kilometers away. So that was a problem.
- 1.37** In addition to that, he was employed, so he had obligations; he was meant to be present and working. But Buxtehude was one of the greatest living exponents of the northern German organ school, a great improviser, a great composer. And Bach was absolutely determined to go and see him and see what he could learn from Buxtehude. So he decides to go off. He obtains leave from his employer, and he's expected back in four weeks.
- 1.38** The problem was that he had no money. So he decides to walk 400 kilometers on foot. Amazingly, he does it. He turns up in Lübeck. Apparently, he gets on well with

Buxtehude, decides to stay a little longer. So he remains for four months and then apparently walks back 400 kilometers. That got him into pretty serious trouble. There was no WhatsApp in those days, so I'm sure his employers were wondering what was going on, and that led to him being fired.

- 1.39 There are similar instances of Bach being reprimanded by his employers throughout his career. He seems to have been a very strong-willed person, constantly at odds with the structures within which he was obligated to operate. He was even briefly jailed at one point for violating the terms of his contract. These positions kept coming, and he kept trying to improve his situation as a musician in order to have the broadest range of musical opportunities possible.
- 1.40 After his nearly 10 years in Weimar, he gets bored and starts looking for something else. And he's named Kapellmeister in Köthen, where he stays for about six years. He comes into another problem in Köthen, which is that the Prince was himself not particularly interested in music. He was also a Calvinist, which meant that there wasn't a great need for the sort of music that Bach was interested in writing.
- 1.41 So he keeps looking, and eventually he's considered for a position in Leipzig, which lies in Saxony, but it's still very relatively close by, still not really leaving this very condensed geographic region. Interestingly, by this point, Bach has produced many of his greatest works. He's well known as one of the great virtuosos. There's no doubt as to his consummate skill and his artistry.
- 1.42 But amazingly, from our present-day perspective, where Bach is universally venerated, played all over the world by pretty much every musician who learns an instrument, when he applies for this position in Leipzig as the Thomaskantor, or the director of music for the Thomasschule and the Thomaskirche, which are a music school and also a church for which a steady production of music was required by its music director, he applies for this position and he's in third place. He's in third place.
- 1.43 The musician they really wanted was Telemann, who, as I mentioned earlier, was this colossally prolific composer, working in every conceivable genre, writing dozens, if not hundreds, of sonatas and concertos and operas, and also an extremely dynamic character, gregarious, charming, well-loved, writing music that was intensely popular and widely circulated during his life. So they wanted him. They didn't necessarily go looking for Bach, but Telemann turned them down and their second choice turned them down.
- 1.44 So one of the city councilors said, "Well, we would have liked to have the best, but we'll settle for a middling option," and they hired Bach. So in 1723, he takes up his new position in Leipzig, and his duties there are astounding to contemplate. He's responsible for the musical education of all the students at the Thomasschule, and he's also responsible for maintaining the instruments. He's responsible for producing astounding quantities of music for all the events of the liturgical calendar—we'll get

into what that specifically means a little bit later. He also has to rehearse the pieces. He has to direct the rehearsals. He has to perform. He has to play organ. It's dizzying.

- 1.45** It's worth pointing out, too, that the Bach family household was a very busy one. Bach was married twice. His first wife died unexpectedly at a relatively young age. In total, between his two wives, they had 20 children. On top of that, Bach was very involved in the musical education of his kids, not only overseeing it, but composing great pieces for them.
- 1.46** So in Leipzig, he throws himself into his work with unbelievable fervor. It's as though some kind of limitation has been removed, and he enthusiastically begins this project of creating pieces of music for all the liturgical occasions. And what that actually meant was writing a complete 20 to 25 minute piece of music for chorus and orchestra every week, on top of everything else he's doing. This is where it starts to get a little bit surreal. It's very, very difficult even for scholars to understand how he did that.
- 1.47** So he completes a full cycle of these pieces. That's about 60 pieces a year, at least for all the Sundays, all the feast days. On top of that, he's also working on his own personal projects. So by the end of the first year, he's completed how many hours of music? 20 to 25 minutes per piece, 60 pieces. And then, the next year, he does it again, a second cycle and a third cycle. So the 1720s for Bach are a period of apparently unending work. This is again, where we come to the aspect of Bach that is maybe difficult to relate to for mere mortals like myself. How did he sustain that at the highest level of musical invention?
- 1.48** I want to show you an example of what I mean. So I said that Bach had the capacity to extemporize on pretty much anything, no matter how seemingly unpromising or uninteresting the musical material was. One of the first times in my life that I really understood what that meant was when I started reading about one of his later works, The Musical Offering, which he composed three years before his death in 1747.
- 1.49** Without going into too much detail about that piece, because we're going to be looking at it more specifically a little bit later, I'll just briefly say that on a visit to Potsdam, he meets the King of Prussia, Frederick the Great, who gives him a musical subject and asks, Bach, "So you're the great Bach, you're this great improviser, this great composer. What can you do with this?" And he gives him this subject. And the subject seems to have been deliberately invented to be as perverse as possible.
- 1.50** It's completely—it's chromatic, it's strange, it's kind of lopsided. So he's like, "What can you do with this?" Bach sits down and improvises a three-part fugue based on this subject. And then, he's asked, "Could you do a six-part fugue?" And he does. Then he goes back home again to Leipzig, reflects on his experience and thinks, "I could have done better than that." So he starts writing a whole collection of pieces based on this theme.

- 1.51** One of the pieces—this got me really excited when I was 18—one of the pieces is the same backwards and forwards. And I found out about this through a book called Gödel, Escher, Bach, written by Douglas Hofstadter. This is a thick but very entertaining book about, among other things, mathematical concepts such as eternal recursion, structures that loop back onto themselves the work of M. C. Escher.
- 1.52** Hofstadter took an interest in the music of Bach, which sometimes has these properties, and he talks about The Musical Offering. He talks about this piece, which is a Canon a 2, a canon in two voices, where, in the published score of the piece, Bach puts this little enigmatic inscription. And he says something like, "If you seek, you will find. You'll find a solution." The piece is presented just as a single line of music with the hint that there's something to be found within it.
- 1.53** There's a whole series of these. There's 10 of them in The Musical Offering, and they're called puzzle canons, because the idea was that you have to figure out what's going on here and figure out a way to combine this line with itself in a way that creates a complete piece. But he doesn't give you the solution. So performers, interested parties could buy this score, look at these fragmentary lines, and try to figure out exactly what he intended.
- 1.54** So I'm going to play you the theme that Frederick the Great gave to Bach [plays the theme on the piano]. The problem with that theme is that it's full of awkward and strange leaps, and it's got this strange chromatic descent [plays the chromatic descent], which, if you have to harmonize every note of that and combine it with itself at different intervals, that's not an easy task.
- 1.55** So what does he do? He writes this canon where if you combine the two voices in a certain way, or combine the line with itself in a certain way, you end up with this piece that is perfectly retrogradable. It's the same in either direction. So when you open the score of The Musical Offering, you see this line. He starts with the subject given to him by Frederick the Great, and then there's a little ornamental passage that follows it, and it sounds like this [plays the single melodic line on the piano].
- 1.56** So here's what happens when you solve the puzzle [plays the completed canon on the piano]. He takes delight in solving impossible problems. When you hear this piece, it sounds like a beautiful piece, but you don't imagine the unbelievable contortions he had to go through in order to make this work technically. That's the thing. The artistry just disappears.
- 1.57** It's one thing to make a spectacle of your virtuosity, to impress your audience with your daredevilry, but in this little piece, all of that's pretty much invisible. And if you're not really paying attention, you might not even notice what's actually happening. Now listen to the mirror point, if you like, the point at which the piece reverses and changes direction. This piece, incidentally, is referred to as a cancrizans or a crab canon, because

of the lateral walking patterns of crabs. And once it reaches the middle point, it flips around and keeps going [continues backward from the midpoint].

- 1.58** Can you hear this? [continues playing as the canon repeats]. You can keep going forever. It keeps repeating itself. And when I heard about this piece and got a hold of the recording and started listening to it and studying it, I was fascinated. I thought, "What kind of a mind does this guy have where he can do these things and it's still intensely expressive, it's still a beautiful piece of music, but he's solving these seemingly impossible technical problems at the same time?" That really impressed me.
- 1.59** There's a quote I'm going to share with you, not from a contemporary of Bach's, but a music critic and composer from the 19th century called Johann Friedrich Reichardt, who writes that Bach, as far as he's concerned, is "the greatest example in any art form of a master's ability to move with freedom and assurance, even in chains." I think one of the reasons that I found this so fascinating is that Bach is one of the great illustrations of what I believe to be one of the most fundamental principles that underlie creativity and artistic invention, which is that the truest freedom comes from limitations. It comes from the voluntary adoption of constraint.
- 1.60** This is a concept that I think we as a culture have maybe not completely internalized. Our attitude towards art has become a kind of adolescent fascination with what is effectively a Romantic trope, a Romantic cliché of the artist obeying nothing but their own internal compass and producing, as it were, snapshots of their fleeting, interior emotional states, something like that. That concept, by the way, never even applied to the Romantics. It's a complete fiction. But we have this idea of the artist as somebody who breaks free of societal constraints and expresses their own freedom, their own subjectivity.
- 1.61** Bach, in a sense, is the opposite of that. He voluntarily submits his art to the most recondite series of constraints you can possibly imagine and flourishes under them somehow. And by constraints, I don't just mean pieces that you can flip backwards and forwards, and they work either way, and you can exchange the voices, and you can put one voice on top of the other, and everything works. Or writing pieces that are mirror images of each other, or God knows what else, all kinds of dizzying, contrapuntal elaborations.
- 1.62** It's the constraints that come from writing pieces that have an intended purpose behind them. It's not art for the sake of art. And what Bach gives us is this multifaceted, astoundingly complete worldview. His music reflects theology, philosophy, science, a form of entertainment. He wrote comic pieces, dance pieces, pedagogy. He was fascinated with new technical possibilities, of instrument building, of new systems, of tuning, of temperament, of keyboard instruments that allowed him to do things like that.

- 1.63** So his music reflects a complete person and a complete worldview in a dizzying inventiveness. It's hard to come up with another way of explaining it. That's really what I mean by the breadth and the depth of his accomplishment. It really does seem to have everything, but on top of that, it's also some of the most human and the most deeply moving music that you can hear.
- 1.64** So that was my introduction to Bach. I went out and got all the recordings I could find of Glenn Gould playing all the keyboard music, and got the scores and tried bashing through them myself, and it conditioned my entire life as a composer effectively.
- 1.65** The thing that I found the most rewarding, personally, the most fascinating, was the fact that in Bach, your ears are always being stretched. He's always trying to see: How much can he give you? How many different lines can go on simultaneously that complement each other and that add up to an extraordinarily rich texture? It was that polyphonic dimension.
- 1.66** The fact that you're always hearing these independent lines where each little line, if you isolate it, as was the case in that little canon, you can just play one of the lines, and you get a perfectly beautiful, musical, interesting line. And then you combine it with itself, and you get a perfectly beautiful, amazing harmony. And you can follow each line independently. There's nothing I can think of that's more musically satisfying than this experience of multiplicity within a unified framework.
- 1.67** In that sense, Bach was not really a man of his time. This is also something that is an important point to consider: the obsession with counterpoint, with polyphony, with the elaboration of multiple independent lines that proceed simultaneously, was, by the time he was advanced in his career, a totally anachronistic idea. The art that Bach was pushing to its limits was really the art of the old polyphonic style, which characterized European music, broadly speaking, for hundreds of years; in other words, writing for multiple voices, each of which has its own independent part.
- 1.68** By the time Bach was relatively young, a massive sea change in music had already taken place. Music had moved away from the old modal polyphonic style, with its independent lines and its rather free-floating combinations, all of which were really predicated on the horizontal dimension of music, which is to say the melodic dimension. The harmony in those pieces arose out of the superposition of different independent lines, but nobody thought in terms of chords as such.
- 1.69** By the early 1600s, that was starting to change radically, especially with the advent of composers like Claudio Monteverdi, who starts writing operas. With opera, you need to have a single voice that's put to the fore and then something to accompany the voice. You can't have eight different voices, all singing different words at the same time with complete independence, because you wouldn't be able to follow the story. You would just get this texture of syllables, which most of the time are not directly comprehensible.

- 1.70** You have to remember that in the old polyphonic style, the intent was not to give you a clearly audible text, as though you're reading a poem. If you think about contemporary songs, pop songs, most of the time you can clearly hear the words, unless you're Bob Dylan, in which case that's just out the window—most of the time you can.
- 1.71** That, in itself, we think of that as a kind of banal thing. Obviously, you want to be able to hear the words, but in the old polyphonic style, what you would do would be to take the text, usually a sacred text, and dissolve it within this musical tapestry, where the music expresses the idea of the text, turns it into a kind of perfume, if you like, something more diffuse, but you're not necessarily expected to hear it; you're not expected to understand every word.
- 1.72** You can't, because in imitative, polyphonic textures, canonic textures, where each line is independent, the words don't all line up, so you'll hear a syllable here, a syllable there. If you know the text already, then you might be able to know what you're hearing. But in opera, in dramatic forms, where you need to have a single voice, you can't do that. It has to be comprehensible. The lines have to be natural-sounding. They have to be relatable to everyday speech.
- 1.73** So there's this immense sea change that happens around 1600. Bach is born in 1685, and while he's perfectly capable of writing pieces that conform to that style, a lot of his music is written in what's referred to as the *stile antico*, the old style, the old polyphonic style that comes to the fore as his career wears on.
- 1.74** So Bach in his time was something of an anachronism. He was writing in a polyphonic style primarily, but with a harmonic language that was the language of the 1700s, as opposed to the old modal style—I'll explain what that means a little bit later, because it's an important point with respect to Bach—but the thing to retain from this is that those two things don't really go together.
- 1.75** And so, strangely enough, although he lived during the Baroque era and was a man of the 1700s, Bach was really writing in a style that was completely outdated and out of fashion, which didn't do his reputation any favors. Many contemporaries of Bach found the music to be turgid, overly complicated, strange, hard to follow, with no clear melody and an excess of art—that was a term that was used against him: "He ruins and disfigures these simple, plain songs through an excess of art."
- 1.76** So that's a little sketch of who Bach was. He was obviously strong-willed enough and—what would you say?—stubborn enough to persist with a style that in many respects was the final flowering, the highest point of development of the old polyphonic style in an era that had already moved on, and he does so while creating a body of work so profound, so rich, so inexhaustible, that musicians all over the world are still devoting their entire lives to it. Amazing.

1.77 We're going to stop here for now and then we're going to look at some compositions of Bach and I'm going to take you through what's going on stylistically and technically and situate them in their historic context. Thank you.

Lecture 2: Bach's Organ Genius

At a glance — A survey of Bach's organ genius, centered on the Weimar years and works like the Orgelbüchlein — how Bach built an unmatched command of counterpoint and chorale at the keyboard.

- 2.1 Out of necessity, this is going to be an incomplete survey of Bach's works with many gaps, unfortunately. There's no way around that—there's simply too much work to cover even in eight hours. There are going to be entire sets of major pieces that I'm not going to be able to address here. I'm not going to be talking about the Brandenburg Concertos, for example, even though I love those pieces. I'm not really going to touch upon The Well-Tempered Clavier, which is an immense set of preludes and fugues by Bach. There's simply too much.
- 2.2 What I'm going to do is I'm going to touch upon works that represent different periods of his creative evolution and that show different facets of Bach as an artist. We're going to start with two things today. We're going to look at a collection of 46 chorale preludes that Bach wrote during his Weimar years. And these pieces represent one of the earlier indications of his capacity to elaborate upon simple musical ideas. As a collection, this set of pieces is known as the Orgelbüchlein or the Little Organ Book. These pieces were written by and large between 1708 and 1717, although he made later additions to the set.
- 2.3 So remember, Bach at this point is in his early 20s, and he's taking up one of his earlier professional posts, and he sets upon one of his characteristically ambitious compositional projects. He has this idea that he's going to set something like 150 different examples of the plainsong in these polyphonic elaborations for organ. The idea being that for any conceivable liturgical function, you have this huge compendium of pieces to choose from.
- 2.4 As it happens, rather than the 150 or so that he had intended to write, he managed to complete 46 of them, one of which he treats in two different manners in this book. The Orgelbüchlein today is one of the staples of the organist's repertoire. Even in its abbreviated form of only 46 pieces, it's still around 80 minutes of music.
- 2.5 This collection of pieces is something of a touchstone for musicians everywhere, not just organists, not just people interested in liturgical music, but figures as unlikely as the French avant-garde composer and conductor, Pierre Boulez, was fascinated by these pieces in the 1950s and included them on concert programs alongside modern works. That's kind of amazing.
- 2.6 Boulez was famously intransigent, and he had very narrow tastes when it came to what he considered to be of an acceptable level of quality in music and was notoriously ungenerous to many of his colleagues. But there were only a few pieces that seemed to pass his test, and the Orgelbüchlein pieces were among them. Very interesting. Why

would an avant-garde composer gravitate to these little short liturgical pieces from 1708?

- 2.7** Well, these pieces are remarkable in a number of ways, but I think the first is that they compress an enormous amount into an incredibly small space. These 46 pieces are short. They're on average about a minute long. Some of them are a little longer, some of them are shorter, but these are very short little pieces. When you're dealing with a minute, when that's your formal framework, you don't have a lot of time for repetition. You don't have a lot of time to state your ideas more than once and develop them.
- 2.8** You're left with this musical moment that unfolds, and then the piece is very rapidly over. It gives you this very condensed time frame, but a vertiginous time frame in which many things appear to be happening at the same time. What happens is that really messes with your sense of time. Even though the pieces are short, you feel like you've stepped into some strange labyrinth where time is just infinitely distended, or rather, you lose all sense of it.
- 2.9** That's something that appeals to a lot of musicians, including, as I've pointed out, avant-garde musicians like Boulez, who are also interested in those kinds of strange dilations and compressions, and expansions of our experience of time. That's one of the things that music can do admirably well, and it's very difficult to accomplish in any other art form. Yes, the piece in chronometric terms might be a minute and 20 seconds long, but the experience of time is completely different from that. That was one of the most profound lessons for me of this music: how you can completely transcend our normal perceptions of time, the inexorable forward march of time. In these pieces, you seem to catch a glimpse of something else, a glimpse of something intemporal.
- 2.10** How does he do that exactly? Because remember, he's starting from the most humble materials. These were short little tunes, many of them composed by either Martin Luther himself or by a series of poets over the centuries. And these tunes had to be simple. They had to be things that anybody could sing. They were intended to be sung in unison in a liturgical setting.
- 2.11** What he does is he takes these melodies and embeds them within this vortex of complicated patterns so that the song itself, which is in a sense the focal point of the piece—it's what gives the piece its title, and it's the most immediately recognizable element of it. But in a way, the melody disappears into this labyrinth and becomes, in a sense, not even the focal point of the piece anymore. It's there, you can hear it, but it's a pretext for Bach to elaborate musically.
- 2.12** So his capacity to compress time, to give us a lived experience, a phenomenology of time that's utterly different than what we would ever experience in normal life, and to do that in a way that is dizzyingly beautiful and completely absorbing, gives you this sensation of limitless multiplicity.

- 2.13** We're going to talk about the Orgelbüchlein. I'm going to play you a couple of examples of it. I'm going to show you how the pieces are put together. We're also going to look at a couple of his four-part, chorale settings. The reason we're starting there is that this allows me to give you a picture of what's really at the center of Bach's musical preoccupations.
- 2.14** If you take any of the pieces, even the most recondite works, the most sophisticated pieces, the ones where he takes his art of counterpoint and elaborates it to the highest degree of sophistication, he's always starting from a little seed. There's something very simple at the heart of all these pieces. A few notes, a simple little hymnsong, and it starts from there and it grows outwards in all of its possible manifold ramifications.
- 2.15** Bach was a profoundly devout man and a profoundly devout composer. One of the central ideas of his whole project is to take things that were familiar to everybody and to deal with very human emotions, not trivial emotions, not fleeting emotions, but the big ones—the heights of tragedy, the heights of jubilation, the most extreme manifestations of human experience—to have that at the center of his preoccupations.
- 2.16** So these little chorale melodies, humble as they are, simple as they are, keep coming up again and again and again in his work like a continuous thread. The chorale preludes of the Orgelbüchlein are polyphonic elaborations in that old style upon these pieces. The four-part settings of chorales, where he takes the melody and adds an additional three voices, harmonizes them for four parts—soprano, alto, tenor, and bass—are chorale textures, which is to say that for the most part, everybody's singing the words essentially in unison, rhythmic unison, rather. It's different ways of addressing the same fundamental concern. Then later on today, when we start looking at his cantatas, you're going to see still other examples of how he treats these materials.
- 2.17** The other thing that we're going to see when we start looking at these pieces in detail is how closely tied up Bach is with the art of rhetoric and musical figuralisms, which is to say that there was a body of symbols, you might say, with which Bach could speak directly to his audience of that time. That's something I fear we may have lost, this idea of a shared symbolic language. It takes a very powerful form in this work, however. If he's talking about sorrow or confusion, a feeling of being lost, a feeling of jubilation, there are musical figures that signal those emotions to us directly without needing words to explain them.
- 2.18** Some of these figuralisms or musical manifestations, musical illustrations of an idea or a symbol, are things that we can pick up instinctively without necessarily articulating why it is we're feeling what we're feeling. Other times, it's a little bit more buried in the piece, and you have to do a bit of digging, maybe to understand the vocabulary of symbols that was used at the time, but they're present in virtually every piece he wrote.

- 2.19** You have to remember that the art of music is an extension of speech, fundamentally. When I pointed out that the center of the European art music tradition, up until about 1600, was this polyphonic style, all of it was predicated on the voice, fundamentally, the individual voice, either alone or in combination with others, most of the time in combination with others. As such, the earliest compositions, the earliest written down pieces, took the form of mnemonic aids. They allowed people to remember scripture or Psalms. It was a way to retain the text more easily if you associate it with a line, an extension of speech.
- 2.20** Each of us, when we speak, we have a cadence to our voice. There's a change in pitch, there's a change in rhythm. Even if you don't understand a language, you can understand when somebody's coming to the end of a point. We can instinctively understand the ends of phrases, the pauses, the changing speed, the different degrees of enunciation and accentuation put on different syllables. All of this is manifest in these pieces as well.
- 2.21** Let's look at one of the 46 chorale preludes from the Orgelbüchlein. Now, this is a piece based around the Christmas carol *In dulci jubilo*, *In Sweet Jubilation*. Now, again, a piece that everybody at the time would have been familiar with. In its original form from 1535, it sounds like this [plays the *In dulci jubilo* melody on the piano].
- 2.22** *In dulci jubilo* from the Orgelbüchlein, Bach takes this piece and turns it into a double canon at the octave—I'm going to explain what that means. So a canon is not a form, it's not a genre, it's something else. It's a process whereby you take a musical line and then play it back upon itself, but staggered in time. At a fixed temporal interval from itself, you repeat the line. You might have a canon at the unison, in which case the pitches are exactly the same, but they're just staggered relative to each other.
- 2.23** You start by singing. It's like a round when singing "row, row, row your boat," where somebody starts and then later the next person comes in and so on. The idea is to strictly stay with a single line, multiply it upon itself, and in a way that is harmonically coherent. That is to say, when the tones are heard together as vertical simultaneities, everything is harmonically coherent.
- 2.24** Now, that's an interesting technical challenge in the harmonic language of Bach's day, because you have to be very careful to make sure that every tone, when played against itself at a fixed temporal interval, is still coherent. You have to make sure that you're not stringing together unresolved dissonances. You have to make sure that the intervals are coherent. In order to do that, when you're writing the line, you have to always think about: How is this note going to combine with itself a bar later? That's tricky enough to do, but it can get trickier.
- 2.25** You can also have a canon at any interval. You can have a canon at the fifth, a canon at the octave, a canon at the seventh, etc., although that's fairly rare. What that means is you take the line, you transpose it so that it's higher, it starts on a higher note, and then

you keep going, but with all the intervals the same. You can also have a canon that repeats itself where the voices are staggered as little as one beat, as much as a bar or even two bars. The possibilities are endless, but the basic principle is always the same, which is the unfolding of a polyphony out of a single voice.

- 2.26** So in this piece, Bach decides to turn *In dulci jubilo* into a double canon at the octave. And what that means is the melody is present in the top voice, and that's called a cantus firmus. A cantus firmus is a line that is heard in the top position in the soprano line around which you can build a piece. You can add supplementary voices to it. But you start with that. It's a fixed melody to which you can add a bass line, an alto, a tenor, or whatever it is. Cantus firmus exercises are part of the training of many composers.
- 2.27** Certainly, in Bach's day, this would have been absolutely standard. You start with a line, you have to harmonize it, or you have to create an accompaniment that goes with it. It's foundational in the art of counterpoint, for example, where the student is given a line and has to write a second line that works around it. So he turns *In dulci jubilo* into a cantus firmus, and he repeats the line, identically, one octave lower, one bar later.
- 2.28** Now, it's hard enough to do a canon where you're writing the line itself because you can work out, "Okay, If I go up, then this happens, and if the note goes down to this note, then that's what happens." But if you're taking a piece that already exists, you can't do that because the melody has to be understandable. You still have to be able to hear it and sing it, and it is what it is. You can't change it. You have to be thinking, "Well, how could I combine this line with itself in a canonical elaboration?"
- 2.29** But he doesn't stop there. He has a canon at the octave, and we'll just look at what that sounds like. Here's the cantus firmus—remember, this is basically the same as the line that I played to you previously. He slightly modifies the rhythm, but only very slightly, and the result is this [plays the *In dulci jubilo* melody on the piano]. And it continues; that's how it begins.
- 2.30** Now, when you take that line and add it to itself one bar later, an octave lower in the left hand, then you get this [plays the melody with a delayed octave entry in the left hand]. Very nice. You already have this impression of multiplicity, of multiple things happening at the same time in different temporal spaces. Bach was fascinated by these formats because they allowed him—he obviously had one of those minds where you need a really hard problem to solve in order to not be bored. This whole book is an amazing demonstration of that.
- 2.31** So that's the first layer in this texture is this canon. But remember, I said this is a double canon at the octave. That means that there's also another layer in the texture that also appears in a canonical relationship with itself. That takes the form of a triplet pattern in the alto voice. It's this flowing rhythm that irrigates the entire piece from beginning to end. It sounds like this [plays the triplet figuration on the piano]. That continues. Of course, since it's a double canon, we also hear it one octave lower, one bar later [plays

the triplet figuration an octave lower], and so on. When you combine the two, the figuration in triplets and the line playing against itself, you end up with a beautiful double canon.

- 2.32** We're going to hear that, and I want you to hear it in an organ version because these are organ pieces. They're intended to be played on an organ with multiple manuals or multiple keyboards with a pedal, a pedal board. The pedal, of course—the pedals are played with the feet. You have to imagine the physical dexterity required to do this. You've got your feet like this, and then you're playing all these multiple voices at the same time on two different keyboards at the same moment. It's incredible. It's called the king of instruments for a reason.
- 2.33** Here's an *In dulci jubilo*. We're going to listen to it. I just want to briefly tell you the text of the original poem so you get an idea of what this expresses: "In quiet joy, let us our homage show. Our heart's joy reclineth in a manger, and like a bright star, shineth in the mother's lap. Thou art Alpha and Omega." It's a Christmas carol. It's a song of jubilation. And those flowing triplets give us that wonderful sense of joy as the piece moves forward. Here's *In dulci jubilo* [recording plays].
- 2.34** I think that's rather good for a man in his early 20s. Now, the *Orgelbüchlein* gives us this whole gamut of human emotions, and that includes jubilation as it includes distress. There's another piece in this collection that had an extraordinarily far-reaching resonance for musicians for hundreds of years. It's a piece called *Durch Adams Fall ist ganz verderbt*, which translates to, "Through the fall of Adam, mankind has become completely corrupted."
- 2.35** Bach, of course, with his preoccupation with musical figuration and finding ways to turn texts into something that you can experience viscerally through sound, takes this as a challenge. The piece that he writes is certainly one of the strangest bits of contrapuntal writing I've ever heard, and also one of the most amazing.
- 2.36** The Austrian composer, Arnold Schönberg, felt that that piece was an extraordinary indication of how far it's possible to push a given language before it just completely collapses. What he does is he takes this tune and he invents a bass line that is completely chromatic. If you hear the bass line—and I'm going to play it for you—but if you hear the bass line on its own, you're going to think that that's wild. It sounds like a piece of atonal music. It sounds completely—the ear can't really relate it to a key. The point of doing it this way is to express a loss of direction, a sensation of confusion and distress.
- 2.37** Let's have a look at how this piece is put together. I'm going to start by reading the text of the hymn. "Through Adam's fall, human nature and character was completely ruined. The same poison has been inherited by us so that we would not be able to recover health without comfort from God, who has redeemed us from the great harm

that was done when the serpent overcame Eve and led her to bring God's wrath upon herself."

- 2.38** Here's what the pedals are doing in this piece on the organ [plays the pedal line on the piano]. What do we notice there? Well, if you hear that on its own and you're trying to figure out—remember, this is written somewhere around 1710, maybe. It has to be in a key. It's impossible to tell what that would be because you have every chromatic pitch, and it never seems to repeat or coalesce around any obvious sense of a tonic or a center.
- 2.39** On top of that, it has all of these wild leaps, these dissonant descending intervals, illustrating the idea of corruption and fall. These are musical figuralisms. So when you hear this [plays the pedal line on the piano], it's a tumbling gesture, a downfall that he's illustrating musically. What's amazing is somehow, harmonically, he makes that work with the melody of the hymn, which sounds like this [plays the melody].
- 2.40** This is another piece written in a four-part texture. What you're going to hear is the cantus firmus, again, in the soprano, the top line. You're going to hear that wild bass line played in the pedals. Then you're going to hear two inner voices that provide this dizzying maze of contrapuntal elaboration that somehow makes those dissonant intervals in the bass with their disjunct profile work together with that top line. Let's listen to *Durch Adams Fall ist ganz verdärbt* [recording plays].
- 2.41** You have to put yourself in the framework of a contemporary organist of that time, where this 22-year-old or whatever comes and writes this thing, which is completely insane. But it's amazing. It's absolutely amazing what he did there. Now, this is why these pieces are so fascinating to composers today. First of all, the willingness to attempt to do these impossible things, to take irreconcilable materials, that extremely dissonant bass line, combine it with this simple plainsong, and then figure out a way to make those things work together in a way that's totally musical and expressive.
- 2.42** Remember that these intervals in the bass, these are really dissonant intervals. He goes from a C to a D $\sharp$ [plays C, followed by D $\sharp$ below it]. That's a diminished seventh, that interval [plays C, followed by D $\sharp$ below it]. He does it again [plays B $\flat$, followed by D $\flat$ below it, then F, followed by G $\flat$ below it]. It's a chain of three diminished sevenths. If you take all of those tones and add them together, you get almost a complete chromatic scale [plays a chromatic scale], every semitone on the keyboard [plays C, followed by D $\sharp$ below it, then B $\flat$, followed by D $\flat$ below it, then F, followed by G $\flat$ below it, then A and D]. He just keeps going from there.
- 2.43** That's two of the 46 pieces in this collection, each of which is a crystalline little world of expressive potential in and of itself. I'm going to play you one final piece: *Vom Himmel kam der Engel Schar*, which translates to *From Heaven Came the Angel*, and this is one of my favorite pieces from this collection [recording plays].

- 2.44 I said that Bach had the capacity to take these plainsongs and present them in a variety of different ways. These, of course, are very sophisticated contrapuntal elaborations. There's a science to that. Counterpoint is a science of figuring out all the different ramifications of combining different voices together. He certainly took that farther than anybody else that you can possibly conceive of.
- 2.45 Now, he also did that in the form of four-part settings—I'm going to play you one. This is a book where, if you've looked at my other courses, you'll know that this is something that I keep returning back to. I've shown it before. This is a collection of 386 odd four-part settings that are to be found in Bach's liturgical pieces in the cantatas, the passions, oratorios. What they are, they're moments where the chorus gets up and sings together, usually with instruments doubling the voices.
- 2.46 He takes these hymns and sets them for four voices. Some of these are very, very simple, and some of them are amazingly elaborate. I'm going to play you one called Christ ist erstanden, Christ is risen. This really shows you, again, his inexhaustible capacity to take simple and humble things and elevate them to an incredible degree. Here's the melody, or at least the beginning of it [plays the melody].
- 2.47 This is a fairly long melody, and it has a characteristic that's actually fairly uncommon in these pieces, which is that it contains a refrain on the words Kyrie eleison, have mercy, which is these repeated Ds [plays D], and what's amazing is every time that little refrain returns in this piece, Bach harmonizes it in a completely different way with a completely different texture. He's always trying to find the greatest possible range of expressions of an idea.
- 2.48 So let me play this piece [plays Christ ist erstanden]. Don't forget that our species can do things like that.
- 2.49 I guess partly what I'm curious about is if there's any sense of how he felt about his own art, if it was this relationship to it where he would write it and forget about it and move on and had this transient relationship to it all as it was going on, or partly the limitation and the impact of the limitation on what's being created, and partly the time restraint—I'm not sure how to phrase my question, but it's to do with, I guess, not being so precious about what you're creating as you're creating it.
- 2.50 I've wondered about that, too. If you're creating that much work all the time, if you're writing it almost faster than musicians can rehearse it, how do you remember what you've done and what's your relationship to that work? But Bach, he was a methodical person. When he was writing his great cycles of cantatas, his cycles of organ preludes, and didactic pieces and so on, it was always with the aim of creating a—there's an encyclopedic dimension to it where he wants to cover all the responsibilities. He wants to aim for this completion. I think he was thinking in those terms.
- 2.51 He seems to have very much remembered what he wrote because there are many, many instances of Bach taking a piece that he'd written 20 years ago and refashioning it

into something else or reusing it at a later point for some other purpose. There are concertos by Bach where he takes movements from cantatas and reconfigures them into an instrumental piece. There are also pieces where he turns a harpsichord concerto into an oboe concerto. There are hymn tunes that he set many times in different ways.

- 2.52 He seems to have kept a close eye on his own production and to have retained certainly not every single note of everything he ever had written, but at least he knew what the inventory was. He knew what he'd done, and he knew what he wanted to do. He pursued that with dogged intensity throughout his life, very methodically and systematically.
- 2.53 But this obligation to produce, to come up with a new cantata every week, sounds like very few people would be up to that task. It sounds superhuman, but in a sense, I want to say we should be so lucky as to have a mission like that, a project, something that pulls you forward as an artist, where it's something external, and you have to rise to the occasion. You have to find a way to do it.
- 2.54 If we don't have that, it's very difficult for creative people to be entirely internally motivated. People think they can do that. They think that they can just will themselves into a situation where they can produce and produce. But usually, there does need to be something coming from outside that pushes you.
- 2.55 He certainly had that. He had an immense degree of conscientiousness as a human being and as an artist. He was very hardworking, very preoccupied with going as far as he could with whatever he was working on. That conscientiousness, combined with the workload that was imposed upon him in Leipzig, combined with the fact that by the 1720s, by 1723, when he embarks upon this cantata cycle, he's by that point getting into his late 30s, he has a wealth of experience behind him, he was up to the task. It's amazing.
- 2.56 But he must have also had a physical constitution that was unassailable because remember that in those days there was no real medicine. Medical treatments were rudimentary at best and often extremely dangerous. He seems to have been lucky in that respect where he wasn't sick very often, and he had extraordinary physical energy to be able to do this.

Lecture 3: Didactic Inventions

At a glance — Andreyev turns to Bach's didactic works — the inventions and teaching pieces — showing how instruction and profound artistry coexist in music written to train hand and mind.

- 3.1 So we've looked at one of the major productions of Bach's Weimar years, which is the *Orgelbüchlein*, and we're going to keep going and look at what he did in his next position, which was as a *Kapellmeister* in the town of Köthen.
- 3.2 So a *Kapellmeister*, which translates to chapel master, was a very prestigious position for composers to hold in Bach's day, and it effectively translated to a director of music for a church, a cathedral, a city. These were positions that were highly sought after, that gave the composer a certain degree of stability and protection. And Bach was fortunate to be given this title in the town of Köthen. While he was there, of course, his duties were somewhat different compared to in Weimar. He was left to his own devices to a greater extent, partly because, as I mentioned earlier, the Prince in Köthen was a Calvinist, and therefore there wasn't that much need for Bach to produce masses of liturgical pieces.
- 3.3 So out of this era of his life comes this new preoccupation that he has with writing monumental, encyclopedic, didactic works. And he gave a title to many of these pieces, which was *Clavier-Übung*, or Keyboard Exercises. And one of the first sets of these pieces that he writes are the *Inventions* and *Sinfonias*. These are pieces for a keyboard written in either two or three voices, so two or three independent parts. And this particular set of pieces was written for his son, Wilhelm Friedemann Bach, as pieces to help with his musical education.
- 3.4 So again, these are ostensibly didactic pieces written for a particular pedagogical purpose. And he succeeded so extraordinarily well in that that these pieces still are taught to virtually every pianist who sits down at the instrument. There's still a model of how you can take the multiple constraints involved in writing pieces simple enough for a relative beginner to play at the keyboard, but that still have enough musical substance and interest that you can listen to them in a concert setting, and you can keep returning to them and finding new musical dimensions to them.
- 3.5 So he starts off by writing a series of what he calls *Inventions*—they don't have that title right away. I believe he called them *Preambulae*, which sort of implies a little prelude or a little sort of informal piece with which you might start a recital, something to warm up the fingers. These are fairly simple two-part pieces, and what's remarkable about them is that in every case, he starts with a little germ of an idea and then manages to spin that into a whole series of adventures, where he'll typically start by establishing the key of the piece. The motif will usually begin in the right hand, and then the left hand will take it up and imitate it, and then there will be this sort of conversational texture between the left and the right hands.

- 3.6** The didactic purpose of that is, among other things, to encourage players at the keyboard to develop independence, rhythmic independence of both hands and an even technique. So the pieces use all five fingers equally, and you have to be able to direct a line with both hands at the same time. He stresses this idea of the equal importance of both hands.
- 3.7** This is, again, something that connects us back with what we were talking about earlier with the style antico, the old polyphonic style, which presupposes that all the voices in a polyphonic texture are of equivalent interest. There's no melodic line that captivates your primary interest, and the rest is all filler. There's no division between a melody and an accompaniment in this style. All the voices are equivalent and equally partake of this conversational texture.
- 3.8** So as you're playing these pieces, you have to be very careful, again, about evenness between the two hands. They're wonderful pieces in that sense. He writes fifteen Inventions, or Preambulae, each of which is in a different key, and that's another thing that was an innovation of Bach at the time.
- 3.9** Now, you remember that the way that keyboard instruments were tuned in 1720, when he was working on these pieces, was not the same as today. Equal temperament is a system of tuning whereby the frequency ratio between any two adjacent tones on the keyboard is the same, the interval is the same size, no matter what semitone you're choosing on the keyboard, which, as I explained in the music theory course, gives you a homogeneous result, allowing you to play in any key on the piano, with everything being equally more or less in tune.
- 3.10** But it comes at a cost, which is that you have to have the intervals be slightly smaller than they would be ideally if the notes were exactly in tune. That just has to do with a physical limitation of tuning systems, which is that you can have perfectly in tune octaves, or you can have every interval be exactly acoustically in tune with itself, but you can't have both at the same time, so you have to choose. And, of course, having octaves on a keyboard that are in tune with themselves is pretty fundamental. So there are various competing systems of tuning keyboards at the time that allowed you to have either exactly perfect fifths or exactly perfect thirds.
- 3.11** But then, if you were to go into remote keys like A $\flat$ or D $\flat$, you would end up with these really funky intonation problems. So that didn't really work very well. And some of the intervals would sound extraordinarily dissonant because of the nature of the compromises necessary to make that tuning system work. Now we don't have to worry about that. You can play in any key; you can be in B $\flat$ or A $\flat$ or D $\flat$ or whatever, and it's all homogeneously in tune.
- 3.12** Bach was taking advantage of the then fairly recent possibilities of temperaments that allowed you to play in many different keys. So he doesn't take you systematically through every possible major and minor key the way he does in the Well-Tempered

Clavier, but he writes fifteen pieces in fifteen different keys, major and minor. And then, later, he adds to that the Three-Part Sinfonias. Those pieces are related to the Inventions, but they're more elaborate. So it's not a matter of simply taking a two-part texture and adding a third voice to it. It's more than that. The forms are more complex, the interactions between the voices are more complex as well, and that allows him to stretch out to a degree and try out some more elaborate textures.

- 3.13** So from this period, when he's Kapellmeister in Köthen, he writes mainly instrumental pieces because again, he wasn't being solicited to write liturgical music. So he writes the Brandenburg Concertos, he writes the first book of the Well-Tempered Clavier, the orchestral suites, the cello suites, which are some of the most popular pieces by Bach, the violin suites, and he writes these Two- and Three-Part Inventions.
- 3.14** So what I'm going to do is I'm going to take you through some of these pieces and show you how they're put together, what some of the technical and aesthetic precepts of them are, and how he takes a little microscopic idea and spins it into a whole piece.
- 3.15** I'm going to start with one of the Two-Part Inventions. Each of these pieces has this little cell of an idea that propagates and expands. Let's look at the third Invention, which is the one in D major. This is the basic motif [plays the opening motif of Bach's Invention in D major]. It's a very simple little flowing line, all in 16th notes, very even rhythm, and you'll notice also that it's entirely in conjunct motion with only a couple of little skips. What that means is, well, first of all, it's fairly straightforward to play because there's not a lot of lateral hand movement on the keyboard, and it also encourages players, keyboardists, to develop equal strength on all the fingers.
- 3.16** Let's hear that again [plays the opening motif again]. You can really do all of that without moving the position of your hand at all. Then that line is taken up in the left hand [plays the same line an octave lower], and so on and so forth. And then, there's a kind of conversation that goes on.
- 3.17** Some of these pieces, not this one, but some of the pieces in this collection, use a contrapuntal technique called invertible counterpoint. Invertible counterpoint—this comes back to Bach's fascination with technical devices that are constricting—it means that there's an upper voice and a lower voice, and you can flip their positions around so that the lower voice is played in the left hand and the upper voice is played in the right hand, and they still work harmonically. In other words, all the intervals can be inverted by flipping the hands around, and you still get a coherent result.
- 3.18** So in some of these pieces, you'll hear a motif in the right hand and then a little accompaniment in the left hand, and then they'll flip around and do exactly the same thing, but in the opposite position. Whether or not the piece uses some form of invertible counterpoint or double counterpoint, all of them have this basically conversational texture. What you're going to notice is that these little motifs are heard in a myriad of different forms. Sometimes they're inverted, sometimes they appear to

be flipping backwards, sometimes they're transposed, but they propagate like little cellular organisms.

- 3.19** So let's hear what this piece sounds like [plays Bach's Invention in D major]. It's a bright little piece written in 3/8, which is a simple dance rhythm, very nice to play. And you might have noticed that there are these little motifs that recur at the end of the piece. It's not exactly a clear-cut ABA type of form, right, where you have an opening section, a contrasting middle section, and then a return at the end. It's not quite that straightforward, but there's a version of that in this piece where he starts by laying out a home base, as it were, establishing a tonic. He stays on it for a while. The idea played in the right hand is taken up by the left hand.
- 3.20** Then, once that's done, he takes the piece on a little walk. Then he starts modulating, he starts moving towards other keys, like B minor, for example. And then there's a period of instability in the piece where he circulates through these different keys, always staying with his basic motifs that he's announced at the outset. And then he comes back to D major at the end, and we hear the same ideas again, but in a slightly different form.
- 3.21** Let's hear how that works [plays the opening motif]. Notice how the left hand is taking up the exact same thing. It's not a strict canon, but it's an imitative texture, as though the hands are almost chasing each other, imitating each other's material [plays the opening motif again]. Now, this part of the piece here, you'll notice that the left hand is doing this simple pattern in octaves [plays octave figure]. That's a simple little dance figure, and it's used to give us a little bit of a feeling of rhythmic propulsion, while the right hand continues to elaborate these flowing 16th note patterns [plays octave figures with the left hand and 16th-note patterns with the right hand].
- 3.22** Now, that idea returns towards the end of the piece, but with the hands flipped around. So the right hand is playing that octave figure [plays octave figure]. And the left hand is playing those flowing 16th note patterns. And the result is this [plays octave figure in the right hand and 16th-note patterns in the left hand]. So everything in the piece pretty much can be permuted and flipped around and played in any combination effectively. So they're wonderful didactic pieces. And they also have this feeling of continuous growth and development based around a single idea.
- 3.23** One of the other things he does in these pieces that makes them singularly enjoyable, I think—and this is a common textual feature of a lot of his pieces—is the principle of rhythmic complementarity. Rhythmic complementarity is not an abstract technical idea, but it's also something that has very real practical consequences for the player. What it basically means is that if I'm playing a pattern in 16th notes in the right hand [demonstrates 16th-note pattern], then the left hand is often only playing 8th notes [demonstrates eighth-note pattern].

- 3.24** What that means is if you add the rhythms of the two parts together, you always end up with a continuous flow of 16th notes that's never really interrupted in the piece. So one hand will play slower rhythmic values, the other hand will be playing faster rhythmic values, and they'll be continuously exchanging, which contributes to this idea of a seamless integration of both parts into a texture that can't be divided up.
- 3.25** So, as you listen to it, notice how you have this continuously flowing rhythm, but most of the time you have one hand playing something a little slower and one hand playing something a little faster. There are some occasions where the hands briefly come together, and they're both playing 16th notes, but he doesn't stay there for very long. And usually when that happens, they're playing at some fixed interval from each other. The voices seem to fuse together like that. He only does that very briefly, where both hands are playing a 16th note rhythm [plays 16th-note patterns in both hands].
- 3.26** I'm going to play a portion of the piece again, and I want you to focus this time on this continuous rhythmic flow and the way that he prevents it from becoming monotonous by breaking up the rhythmic figures between the left and the right hands [plays Bach's Invention in D major again].
- 3.27** Now, there's another thing I want to point out here, which is that he has these little lines emerging out of the bass, whereas the right hand is just holding a note for a few bars. What that does also is it periodically focuses you on just one line or just the other. It's as though there's a little solo just for the left hand. So, for example [plays the passage with alternating lines between the hands on the piano]. That's where that conversational texture that's so important in these pieces really becomes even more explicit.
- 3.28** If you were to isolate just the left hand and hear it on its own [plays the left-hand line only], every line is this complete statement. And then when you add the right hand [plays the right-hand line only], and the two hands together [plays the passage with alternating lines between the hands on the piano together]. It gives you this beautiful braided-together texture of two equal voices. So those are the Two-Part Inventions.
- 3.29** Another thing that I find remarkable about these pieces is that the way that Bach was working with his material and the way he thought of his whole mission as a musician maps remarkably well onto the different facets of classical rhetoric devised by Cicero and Aristotle 400 years before him. If you read about classical rhetoric, you learn about the different disciplines that go along with that, including Inventio.
- 3.30** Inventio has to do with first determining: What is the area of investigation? What is the subject that we're going to be dealing with when we give a talk? This is all broken down into component parts so that you can master the art of classical rhetoric. And what I find amazing is when Bach published the final versions of these pieces—remember, they were originally called Preambulae, at least the Two-Part Inventions. The Sinfonias

were originally called Fantasias. When he publishes them, he changes the title to *Inventio*, 1, 2, 3, 4, et cetera. So there's fifteen of these.

- 3.31** And that's remarkable because it ties in certainly with that idea of classical rhetoric, where the idea is you start by choosing your subject, and the subject in this case is that little cell, that little pattern that you hear at the beginning. And then the next part is *Elaboratio*, where you elaborate upon the idea and you take it on a walk, as it were, and show all the different ways you can combine it. The final part is *Executio*, where you execute the idea. And that has to do with not only his capacity as a performer, his virtuosity on the keyboard, but also his ability to improvise.
- 3.32** Let's look at how this manifests in the Three-Part Sinfonias. Remember, I said these pieces are not just Inventions with an additional third voice, but rather they're cooked up in a very different manner. What I find also remarkable in this piece here that I'm about to play is that there's likewise this very simple idea that you hear right at the very beginning, a very short little phrase, and it gets passed back and forth between all three voices this time. The middle voice is sometimes played with the right hand, sometimes with the left, and sometimes it gets passed back and forth. So it's more complex than the Two-Part Inventions, certainly.
- 3.33** These pieces are also more far-reaching in their use of chromaticism and in Bach's willingness to depart from the original key. So they trace a very interesting trajectory. I'll show you what the basic idea is in this piece.
- 3.34** I'm going to play you just the top voice of the three that you hear in the right hand [plays the top voice]. What you just heard was a simple little arpeggio [plays the arpeggiated figure], just the notes of a chord smeared out over time as it were, followed by a little descent [plays the descending figure on the piano]. If you follow that descent all the way to the end of the phrase, it's a descending minor scale [plays the descending minor scale]. So the predominant mode of melodic elaboration in this piece is the descent, which is in keeping with the minor key. Those two things tend to go together expressively.
- 3.35** So let's hear this. G minor Sinfonia Number 11 [plays Bach's Sinfonia in G minor]. So, like the D major Invention in two parts that we heard, this piece is also in 3/8, and it also has this characteristic flow of 16th notes. But in this case, it's passed back and forth between three different voices.
- 3.36** One of the things that I find really beautiful in this piece, and that I often mention to my composition students, is how the distance between the hands, the distance between the voices, the intervals between them, is constantly changing. It's a dynamic variable in the piece where sometimes the hands are like this—all three voices are clustered together into a very small space—and sometimes it opens up like that.
- 3.37** You're not necessarily focusing on that when you're listening, but the piece is constantly moving around in terms of its register from the center of the keyboard

outwards, downwards, parallel motion, contrary motion, back towards the center. It's like watching two crabs crawl over the keyboard. That gives it this wonderful feeling of continuous motion. And sometimes that results in these dissonances where you have two very closely related voices sort of chasing each other.

- 3.38** I'm going to just play the top voice and the middle voice of the opening, and maybe you'll see what I mean [plays the top line and the middle line of the opening section]. The lower of the two voices is imitating once again what the upper voice is doing [plays the middle line only]. And when you add them together, you get this very close braiding together [plays the top line and the middle line of the opening section again]. Now let's try it with all three voices [plays all three lines together], and so on.
- 3.39** The point of pulling apart these different strands is to get accustomed to listening polyphonically, to hearing all of these different independent lines in the music and how they fit together. So we've spoken about all these different techniques that Bach uses to elaborate these pieces, about invertible counterpoint and rhythmic complementarity, and the idea of starting out by laying out his idea, his *Inventio*, this little motif, a period of wandering where it seems to move around into lots of different places, then the harmony becomes unstable and then a return at the end.
- 3.40** This again is a kind of embryo of the dialectical forms of the Classical era, particularly sonata form, where you have a dialectical interaction between two contrasting musical ideas. That doesn't exactly happen here, because there isn't really any contrast between an idea A and an idea B. But you do have this kind of microcosm of a narrative idea, where you establish a home base, you wander, you move away from it, and then you return back to it again at the end. So these little pieces contain quite a lot.
- 3.41** I'm going to play one more. This is another of the *Sinfonias*, and this one is in D minor; it's number four. This is a piece I've always particularly loved to play. I'll start by showing you what the motif is in the right hand [plays the top line first, then all three lines together].
- 3.42** So hopefully, by this point, that basic process which occurs in all of these pieces, the Two-Part and the Three-Part Inventions and *Sinphonias*, is clear. Hopefully, you can notice that basic trajectory of establishing an idea, establishing a starting point, wandering, moving around, exchanging all the voices, and then coming to a close at the end, back where he started again.
- 3.43** I can't possibly give you an overview of all the work that was produced during this period, because there's so much of it, but if you have a chance, you can listen to the *Brandenburg Concertos*, which are suites of contrasting movements, extremely beautiful pieces. You can listen to the cello suites, which are very famous. There's so much wonderful, purely instrumental music from this era.
- 3.44** But as that period of his life wore on, Bach became restless, wanted to engage more again with vocal music, with liturgical music, and that drove him to seek the position in

Leipzig. In the next lecture, we're going to look at what he did while he was there. So I'm going to stop there with the Inventions and Sinfonias, and I'm happy to take any questions that you might have.

- 3.45** I was wondering about the techniques, such as invertible counterpoint and the wandering of the ideas, the musical ideas: Are they unique to Bach during his time, or is he just really good at using those techniques personally? And also, when he's pushing a musical idea to its limit, do you think he does that more intuitively, or is he following a formula? Is it more calculated?
- 3.46** Well, again, Bach appears to have had this combinatorial mind where he could come up with a little motif and immediately imagine he could just run rapidly through all the different possibilities of combining it with itself, backwards with itself, inverted with itself, in rhythmic augmentation and diminution, which is another contrapuntal process where you take a line and play it at half speed or at double speed and then combine it with itself.
- 3.47** He seems to have had a capacity to do that very rapidly, because if you were attempting to do that in the way that he did it, you might actually have to sit down and work out all those combinations and think, "Well, if I do this, does it work? If I have a canon and it's three beats later, does that work?" And you'd have to actually sit and write it down. But it seems that he didn't have to write it down, or if he did, he didn't keep any of his sketches. But considering the amount of work that he produced, it's hard to imagine that it would have taken him hours and hours to work these things out on paper. I don't get the impression.
- 3.48** So, yeah, I mean, he seems to have had the most extraordinary facility to invent an idea, simultaneously understanding all of the ways that you could treat that idea. As far as whether or not these were standard techniques, they were. So any composer would have studied the arcane recesses of counterpoint, including invertible counterpoint and canons and fugues, and all of these different processes that counterpoint allows composers to elaborate upon. So, to a degree, these were relatively standardized processes and forms and techniques, but he pushed it farther, he pushed it farther.
- 3.49** He simply had that capacity to think ten moves ahead whenever he was doing anything, which is quite amazing because when you're a student, and you're trying to learn to do some of these things. If you want to write a fugue, for example, or you want to write that piece of two-part counterpoint, or you want to harmonize a chorale, if you don't have the capacity to think ahead, think about all the ramifications of every decision you're making as you do it—if I put this note here against this other one, then what does that mean about the next few beats? Where are the hands going to go from there? Where do the voices go?
- 3.50** You have to be able to think ahead, and that's quite hard to do. Students will get themselves into trouble all the time by not doing that. And then they get stuck, and

there's no solution. So obviously, he had an extraordinarily fast mind that could just run through all of those things and then pick the best one. That's as best as I can understand it.

- 3.51** My question is about the technique of rhythm that you were talking about, in particular, the divergence in rhythm between the two hands on the keyboard, and that it would occasionally converge again: To what extent does that contribute to the feeling of timelessness that's created by Bach's music? You mentioned in an earlier lecture that time can typically seem compressed, or you lose track of time as you're listening to the music. And I just wondered if that he was using the technique of rhythm there to create that effect.
- 3.52** What I think it does is it creates a remarkably unified texture. Texture is a concept that's really important in musical composition, but that's sometimes difficult to get a grasp on. What it basically means is that if you take all of the elements that are in the piece—that could be a melodic line plus an accompaniment, or it could be three independent voices, or it could be even just a single line or whatever it is—there's a certain homogeneity of the way that those elements are laid out textually.
- 3.53** For example, a two-part texture is one type of texture, a three-part texture, or a monophonic texture, or a melody and accompaniment texture, and so on—there's all these different possibilities. With Bach, what you get are these voices that simultaneously have this complete independence. I've tried to demonstrate that on the keyboard by pointing out that you can isolate any one of the three parts and you get this coherent, beautiful line, but then when you add them together, they also work seamlessly. So you get both an overall homogeneity of texture, where everything is beautifully blended together, but without sacrificing the individuality of each line.
- 3.54** That seems to have been an ideal that he was working towards. There are many ways that he achieves that technically, but one of them, I believe, is that principle of rhythmic complementarity. What that does is it prevents any one particular voice from dominating rhythmically, because your ear will tend to be drawn to the voice that's playing the fastest figurations. But by continuously switching it up between the voices, it mitigates the possibility that one of the voices will dominate.
- 3.55** It's an amazing thing to listen to these polyphonic compositions because your ear is going to wander all over the place. It's fascinating, from a phenomenological point of view, to observe your own process of listening while that's happening. What are you actually hearing? Are you really hearing each voice as it's unfolding? Are you hearing a global texture, a global overall effect, but without necessarily thinking about the individuality of the lines? Or are you sort of hopping from one line to the next? And sometimes you're thinking about the bass, sometimes you're thinking about the middle voice, and it keeps switching up.

- 3.56** I think the answer is it's different for everybody, and it's different each time you listen to a piece like this, which is one of the things that makes them so extraordinarily absorbing. It's one of the reasons that you can listen to them 600 times and not get tired of them.
- 3.57** So what it comes down to is his preoccupation, again, with making sure that every line—if you were thinking in terms of polyphony, you'd be thinking in terms of individual singers. You don't want anyone to have a boring or awkward line. You want every singer to have some important role to play in bringing about this beautiful texture. Even when he's writing for keyboard, he's still thinking that way. He's thinking in terms of voices. So that's a big part of how he's able to start, I believe, from these very, very simple little germinal cells, and even in the simplest little pieces, to make something totally absorbing.
- 3.58** What do you think that Bach would think of the current musical landscape? What would he think about Skrillex, for instance?
- 3.59** Skrillex? Well, you have to remember, Bach never ventured very far outside of central Germany—Thuringia and Saxony were his areas of professional activity. That being said, by the standards of his day, Bach was actually quite interested in what was happening in different parts of Europe, and he was familiar with it. He was familiar with the music of his colleagues and contemporaries in Germany. He was familiar with the preceding generations, but he also knew about what was happening in Italy and what was happening in France and what was happening in England. When you look at his body of works, you're going to see pieces that are influenced by all of those styles.
- 3.60** For example, he wrote Italian concertos in the style of Vivaldi. He also wrote French suites, and he wrote English suites. All of these styles contributed to his own approach to composition. So he's pulling in influences from lots of different parts of the world at that time, and that in itself is a fairly remarkable thing. So he's always looking at what other people are doing, paying attention to it, and seeing if he can learn something from it and integrate that into his own work.
- 3.61** The big difference between the 1720s and today—well, the big difference, there's hundreds of differences, obviously—but one major one is that at that time, the tools and techniques of composition as it was taught, and the treatises and the harmonic textbooks and textbooks on figured bass, and in counterpoint, all of these different techniques, were pretty consistent from one country to another.
- 3.62** There were certainly major stylistic differences between Italy and France, for example, but there was a commonality of technique nevertheless, even though the styles diverted and the expressive aims were not the same. Nevertheless, you could listen to a piece by Scarlatti or Couperin or Bach or Mozart much later, and you wouldn't be completely lost. You wouldn't feel like, "Oh, I have no idea what's going on here

musically." It would be a different style, it wouldn't have the same priorities, but you would understand how it worked technically.

- 3.63** Contemporary historians and theorists have taken to referring to that whole era and beyond as the common practice period; in other words, the point where there was a kind of unity of basic technique amongst musicians in different countries. Whatever we're in now, we're not in a common practice period; we're in the opposite of that. We're in an era of technical and stylistic fragmentation into micro genres and micro styles and subcultures and sub-sub-subcultures.
- 3.64** What I find fascinating is that even in the 70s and 80s, there were artists, where everybody knew who they were, everybody listened to them. It didn't matter if you were a cab driver or an academic working at Oxford. Everybody knew the Beatles. Everybody knew Michael Jackson. You had these cultural phenomena that just completely transcended those barriers.
- 3.65** We don't have that anymore. We have extremely popular pop artists, but it's like they have a much more specific audience now. Everything's a niche. Even Taylor Swift is a niche. She's popular with 12-year-old girls. I don't think she's that popular with Oxford academics—I could be wrong. We could run a poll and find out if that's true or not.
- 3.66** But the point is, it's getting harder and harder to find cultural actors, artists producing work that can work through all of these different microscopic bubbles that we're all in, and that seems to be intensifying with social media, and it's certainly intensifying with the fragmentation of audiences. So along with that comes a diversity of style and technique, but also of symbols, and we don't really have that.
- 3.67** So I think if Bach were alive today, he would, being who he was and being from where he was from, he would be completely incapable of comprehending our current situation. His world was a small world. It was a very unified world. It was one that had clear hierarchies and a clear center to it. The center for him, of course, was religion and the church. And I think if you were witnessing the explosion of virtually infinite stylistic possibilities today that seem to be spinning apart from each other without any center at all, I think he would be mystified and perplexed.
- 3.68** So we're all used to that because that's our existential condition, being alive in 2025, but from his perspective. There are still pockets of the world where people don't travel very much, and they're very focused on a particular community and way of doing things. But a lot of us in the Western world are used to going from one country to another all the time. From an aesthetic point of view and an artistic point of view, anything goes.

Lecture 4: Cantatas

At a glance — An examination of the cantatas Bach produced in Leipzig, their liturgical role, structure, and the fusion of text, chorale, and instrumental writing.

- 4.1 So today we're going to look at one of Bach's cantatas. Now, when I was laying out my biographical sketch of Bach, I mentioned that when he moves to Leipzig, takes his position there as the Thomaskantor, the director of the Thomasschule and the Thomaskirche in Leipzig, he is suddenly given certainly extraordinary musical opportunities, but also burdened with a very heavy load of work.
- 4.2 One of the tasks that he enthusiastically throws himself into is the composition of music to accompany the full calendar of liturgical occasions. That means writing a complete cantata every week for all the Sundays, all the feast days, and a number of other occasions as well. On top of that, he has the ambition of writing complete Passion cycles illustrating the Passion stories. He completes at least two mammoth undertakings, which are the St. Matthew Passion and the St. John Passion. There seems to have been a third one as well—I believe it was St. Luke that doesn't exist in its complete form, but we'll leave that aside for now.
- 4.3 What he did complete were multiple cycles of cantatas starting in 1723. These were not the first cantatas that Bach had written. Already when he was a young man living in Mühlhausen, and later on in Köthen and Leipzig, he was already writing cantatas—some of them were secular cantatas, some of them were sacred cantatas. The early ones that he wrote, which include works such as the Actus Tragicus, or as it's officially known, Gotteszeit ist die allerbeste Zeit, an extraordinary, beautiful piece written in Mühlhausen, or Christ lag in Todesbanden, Christ Lay in the Bonds of Death. These are early cantatas that he wrote.
- 4.4 When you listen to them, you're struck by how unusual they are, formally speaking. That, I believe, is something that he got from Buxtehude, the organist that he went to visit in Lübeck. If you listen to Buxtehude's cantatas, you'll notice some similar qualities, which is that there's no two that are alike. They have very idiosyncratic forms in terms of the types of movements that you see within them. Some of them are passacaglia forms, where there's just a bass line repeating over and over again in a loop. Some of them are written in multiple movements. It was a terrain of experimentation, and there was no fixed format for them.
- 4.5 The early Bach cantatas are like that, where in Christ lag in Todesbanden, for example, he sets an entire hymn, all the verses of it, but each stanza is in a different movement, and the movements are all very, very different from each other musically. Actus Tragicus is another very, very interesting early cantata, where it consists of a large number of very short, little movements that tell a story, but with an almost operatic shift of mood that happens very rapidly over the course of the piece.

- 4.6 By the time he's in Leipzig and completely ensconced in the life of the city, his cantata cycles start to take on somewhat more homogenous characteristics, which is not to say that they're all the same. They certainly aren't all the same. The cycles are clearly different from each other. They have different points of emphasis.
- 4.7 For example, the second complete cycle of cantatas that he wrote are the so-called chorale cantatas, one of which is the piece we're going to look at today. In the chorale cantatas, he takes, once again, one of these Lutheran songs and weaves it throughout the entire piece from beginning to end, more or less. But nevertheless, these pieces, broadly speaking, start to coalesce into something relatively regular, which is that he starts writing pieces consisting of alternating choruses, recitatives, arias, and chorales.
- 4.8 Now, these pieces have one feature in common, which is that they're intended to accompany and illustrate the day's sermon. Bach would take libretti, texts that were written by contemporaries of his, poets—and of course, there had to be an industrial-level production of these texts as well because he had to write a new one every single week—and the libretti would take hymns that typically predated the cantatas by sometimes multiple centuries and build a little story around them with all these little different movements.
- 4.9 What the stories do in these cantatas, effectively, is very interesting, he starts out by laying out a problem of some kind. The problem will be something that relates to the sermon, but that also is relatable to everybody—existential problems, essentially. What do you do about your propensity to sin? What do you do about the fact that you're in some manner or other irredeemably broken? What do you do about mortality? What do you do about confronting the certitude of your own death? What do you do about weakness and disease and all the things that can happen to you that will bring suffering your way?
- 4.10 So he starts out with these archetypal human problems that everyone has to deal with, and then along the way, there are movements that reflect upon this. What happens is that the cantatas are framed more or less as a story, typically with some uplifting arc to it. So in the second half, he then shows you that this is your human condition. You can't do anything about the fact that you're going to have to encounter these problems. But here's what you can do. Here's how you can live your life. Here's how you can respond to these inevitable problems.
- 4.11 A lot of the cantatas have this two-part articulation, where you start with a relatable problem, you meditate on the problem, and then he shows you a solution of some kind, or at least as a way to face it. So these different movement types that he runs through in the cantatas—the choruses, arias, and so on—allow him to show multiple different musical facets within a relatively compressed space.
- 4.12 The cantata we're going to look at today is called *Jesu, der du meine Seele*. This piece, again, typically lays out a fundamental aspect of the human condition, shows you what

choices you have, shows you some ideal that you can aspire towards, such as heaven, for example, and then it focuses on the real world and what you can do.

- 4.13** These cantatas, again, have a number of common features. One of them is that it'll feature a chorus, so typically an S, A, T, B chorus, a mixed chorus of both female and male voices: soprano, alto, tenor, and bass. From the chorus will be drawn soloists. The soloists will sing arias in which you're focusing on a single human individual as opposed to a mass of people. Those are often moments of extraordinary intensity and reflection and humanity within these cycles.
- 4.14** Often, the arias consist of a solo human voice plus an obligato instrument. The obligato is an instrument that accompanies the voice and dovetails with it. So sometimes the obligato instrument, which could be an oboe, it's frequently an oboe in these cantatas, or an oboe d'amore, which is a related, slightly lower-pitched instrument, the love oboe. It could be a violin, it could be a flute, and it plays this duet with the voice, and there's a dance that happens between the two of them.
- 4.15** What's happening in these movements often is that the obligato instrument takes over when the voice can no longer express something, where words are no longer sufficient, then the instrument will take over. The instrument, again, is an extension of the voice in these pieces. The arias will typically have some instrumental texture, usually with what's called continuo. Continuo is a particularly baroque feature that you don't really see in the Classical era; it starts to disappear. By the time Bach was at the end of his life, it was already thoroughly out of fashion.
- 4.16** What continuo is, is it's a bass line above which you see these little numbers, these figures. It's also called figured bass. Keyboardists of the time knew how to read this notation where you would see a D [plays D] in the bass, and then there'd be a little number 3 above it, a little numeral, or a 6, 4, or a 7, or sometimes a sharp sign. There are all these little symbols and little notations. Based on the numbers that appeared above the bass note, you would know what chord to play.
- 4.17** So you had to be educated in the art of realizing figured bass in order to play these parts. What that meant in terms of the texture of the piece was that you would typically have three players realizing a figured bass. There'd be the bass instrument, there would be a polyphonic instrument that was able to play the chords, and there might be a second bass instrument doubling the other one. In principle, you could do it with two, but it was often realized by three players: a bass instrument, a polyphonic instrument, and possibly a third instrument to double them—that could be a bassoon plus a harpsichord, or a double bass plus a lute or a theorbo or any one of these fretted guitar-like instruments that they had in the baroque.
- 4.18** You look at the continuo part, and it appears to be just a single line, but in fact, it has to be realized polyphonically by the players: some continuo, usually some string instruments, and often a solo instrument like an oboe or flute or on top of the voice.

These were not simple pieces. They're elaborate, fully worked-out compositions. There again, when you contemplate the sheer amount of work that goes into writing one of them, it's really quite dizzying.

- 4.19** So we talked about the arias with their figured bass. We talked about the chorales previously—chorales are pretty straightforward to explain; everybody stands up and sings together in four parts. The chorales would typically be featured, at least in the chorale cantatas, at the beginning, where Bach would create a chorale fantasia or fantasy, where the melody of the chorale is woven into a complicated, very elaborate texture involving the orchestra, the soloists, the chorus, and so on. And then it tends to reappear at the end of the piece in the form of a simple four-part chorale with instruments doubling the voices.
- 4.20** The recitative is something that is inherited, among other things, from operatic traditions. What the recitative is, is a solo voice getting up to sing with only a very sparse accompaniment. The accompaniment intentionally does not get in the way of the voice. You really want to be focusing on what the singer is saying. You want it to be comprehensible. Remember that all of these pieces, Bach was a Lutheran, they were all written and sung in vernacular German, so that everybody could understand it. There was no Latin in these pieces. The text had to be comprehensible, and you just had this sparse accompaniment.
- 4.21** Those recitatives where there's just the focus on the one voice are an extension of speech, really. It's an aestheticized, notated speech. Typically, these are either moments where the story advances somehow or another, or sometimes they're more reflective in nature. So chorales, arias, choruses, and recitatives.
- 4.22** Let's listen to what all of this actually sounds like. We're going to listen to the first movement of *Jesu, der du meine Seele*, which is based on a hymn that was written by Johann Rist, R-I-S-T, in 1641. This piece forms a part of his second cycle of Leipzig Cantatas. Let's first look at the melody of the hymn, and I'll read you the words.
- 4.23** Many of these hymns would have multiple verses, but Bach typically didn't set them all. He would set maybe the first one and sometimes the last one [plays the melody line of Bach's *Jesu, der du meine Seele*].
- 4.24** See that they have this rhythm of short little phrases punctuated by pauses. The pauses correspond to the end of the line in each verse. The original German sounds like this:
"Jesu, der du meine Seele, hast durch deinen bitteren Tod aus des Teufels finstern Höhle, und der schweren Seelennot, kräftiglich herausgerissen, und mich solches lassen wissen durch dein angenehmes Wort; sei doch jetzt, o Gott, mein Hort."
- 4.25** "Jesus, you who through your bitter death have with strength, torn my soul from the devil's dark cave and from the heavy anguish of the soul, and have made me to know this through your propitious word, be even now, O God, my refuge."

- 4.26 So let's listen now to how he weaves that melody into the opening movement, the chorale fantasia of his cantata, *Jesu der du meine Seele* [recording plays].
- 4.27 You might have noticed, for those of you who were here for the composition course, there's a device at the end of that piece called a Picardy third, which involves taking a piece that is in a minor key, such as this one, which starts in G minor, and ending it with a major triad right at the end. We start like this [plays a minor chord], and then we end like this [plays a major chord]. Right at the end, it suddenly switches to a major chord.
- 4.28 One of the things that, for me as a composer, is very interesting in this piece is the fact that the entire thing is built around a repeating bass line. It's just a little four-bar phrase in 3/4 that moves from one voice to the next. It starts off in the basses, then it migrates into the middle voices and ends up in the sopranos and moves all over. But it's an obsessively repeating little four-bar phrase that sounds like this [plays the repeating bass line].
- 4.29 At various times, he even inverts it. There are points where you hear it backwards, but it's there pretty much throughout the entire piece. Everything that's happening around it is, in some manner or another, derived harmonically from this bass line.
- 4.30 This bass line contains one of these examples of figuralisms or musical rhetoric that signal to the listener what the fundamental affect of the piece is. The doctrine of the affects was a Baroque concept that had to do with the essential different emotional states that could be expressed using artistic devices.
- 4.31 In terms of music, this piece is a very good example of that because as you move between the different movements of the cantata, you get these wild shifts of affect from this very somber opening, which has an almost lugubrious quality to it, to the second movement, which we'll hear shortly, which is absolutely sprightly and joyous in comparison. So you get these radical shifts of mood from one movement of the piece to the next.
- 4.32 The way that manifests musically here is in the form of a device that was well-known in the Baroque, which is a chromatic fourth. It's a descending motion where you play every semitone between the top note and the fourth below it [plays descending chromatic fourth], and you build a little line around that. In the case of this piece, it takes the form of this very heavy and tragic atmosphere.
- 4.33 I'm going to suggest that we listen now to a recitative from this piece so you can get a sense of what those sound like. They're usually very short, often just a minute or less, and it gives you a chance, first of all, to hear what a continuo part sounds like with the harpsichord and the bass instrument, the harpsichord filling out the harmonies and the bass giving you the basic line. Then, in this case, the tenor is singing the recitative.
- 4.34 The text here is, "Ach, ich bin ein Kind der Sünden," which translates to, "Alas, I am a child of sin." Here we come to the text where there's a meditation on the basic theme

of the piece [recording plays]. The theme of this cantata is very simple. It's just the redemption of sinful, corrupted humanity through the sacrifice of Christ. Of course, that would have related directly to the sermon that was given that day.

- 4.35 It's important, maybe, also to point out that a lot of the texts for these cantatas were written by really second-rate poets. The poetry is actually quite clunky a lot of the time. Some of these texts are not very good at all. It's funny because if you look at the tradition of German lieder or art song, some of the greatest composers in that tradition were setting poems by really second-rate authors. It's not necessarily always first-rate poetry.
- 4.36 I think part of the reason for that might be that if you're dealing with a spectacularly beautiful text, you might not necessarily want to set it to music. You might want to just read it. It could be—this is my own hypothesis, and I might be completely wrong about this—but it could be that working with average texts made it easier to imagine a musical setting without having to worry so much about the integrity of the original text itself.
- 4.37 You can see that in the next piece that I'm going to play you from this cantata, which is a duet for soprano and alto, which is one of my favorite pieces by Bach. I think this is such a wonderful little movement. This piece is called *Wir eilen mit schwachen doch emsigen Schritten*, which translates to *We Hasten With Weak Yet Eager Steps*.
- 4.38 So again, this is really not a first-rate text. I'm going to read you the German original and then the English translation so you can get a sense of it. *Wir eilen mit schwachen, doch emsigen Schritten, o Jesu, o Meister, zu helfen zu Dir. Du suchest die Kranken und Irrenden treulich. Ach höre, wie wir Die Stimmen erheben, um Hülfe zu bitten! Es sei uns dein gnädiges Antlitz erfreulich!*
- 4.39 And that translates to, "We hasten with weak yet eager steps. Oh Jesus, oh Master, to you for help. You faithfully seek the ill and erring. Ah, hear how we lift up our voices too big for help. Let your gracious countenance be joyful to us."
- 4.40 So what you're going to find in this piece is that there's an ostinato pattern in the cello part, which is a simple little rhythmic figure that repeats over and over and over again, like in the opening, chorale fantasia in the piece. The other feature that the piece has is that it's a ritornello. The ritornello is a term derived from the Italian style, particularly concertos, where you have a refrain that keeps recurring over the course of a piece and little episodes in between them.
- 4.41 Bach patterns this as a ritornello and also has this simple little figure, and this figure wonderfully illustrates this idea of hastening forward with these rapid and eager footsteps. So he comes up with this little pattern here. [plays the rapid rhythmic figure]. It's this simple little rhythmic idea, again, a form of figuralism that illustrates the idea behind the text. Let's listen to *Wir eilen mit schwachen doch emsigen Schritten* [recording plays].

- 4.42 The contrast in terms of expression between the recitative that follows this movement and the chorus that precedes it is pretty radical. I mean, this piece sounds positively giddy in comparison to the others. It's amazing to me how he's able to encapsulate all of that within a unified narrative, all based around a single idea.
- 4.43 That's all in a week's work, right? We've heard three movements of a seven-movement cantata. Every movement we've heard so far is a little jewel. They're amazingly well-worked-out, well-thought-out pieces with a completely absorbing continuity. Interesting, distinctive ideas. Nobody else of his time would have been able to write pieces, I think, of this nature and of this caliber in that quantity.
- 4.44 What's fascinating to me, listening to them—and I can't claim to have heard all the cantatas; he wrote well over 300 of them—the inspiration never seems to flag. That's what's so strange. You can listen to any of them, and there's going to be moments of absolutely extraordinary artistic inspiration. Some of them are more famous than others for one reason or another, but they're all of an extraordinarily high standard.
- 4.45 Why did he write so many? Three hundred plus cantatas. This is a question I've been wondering about recently. He fulfilled his essential mission by the end of his first year in Leipzig. He'd written, I think, about sixty cantatas in his initial cycle, and that you would think would be enough; that would fill the repertoire for years to come. But Bach seems to have thought, "Well, maybe the attendees at the Tomaskirche in Leipzig aren't going to want to hear the same piece over and over again, year after year after year. So maybe it would be a good idea to have two cycles, and then you can mix things up a little bit. And then a third cycle."
- 4.46 The third cycle, to be fair, was not a complete cycle written in one year. He spent several years after that filling in the gaps and adding additional cantatas. But nevertheless, this has to be a unique phenomenon in the history of music that he was able to do that and to sustain that level of productivity for such a long time.
- 4.47 Remember, this is only at the halfway point of his life and career. When he's working on this initial cycle, these first two cycles of cantatas in 1723 and 1724, he's—what?—in his late 30s? So this is the midpoint, and he kept going from there into amazing new heights with every new piece.
- 4.48 In trying to understand exactly how he was able to do this and trying to work it out, he must have had a process that simplified the task, at least to a degree. There are some clues. You can look at this bass line [plays the bass line]. You can boil that down into something a lot simpler if you look at what the skeleton of the line is. So a lot of what's going on in these pieces, technically, is that there's a simple stepwise progression that is ornamented through figuration.
- 4.49 For example, if you boil this bass line down to its simplest expression, you end up with something like this [plays the simplified stepwise bass line]. You can come up with a distilled version of the same line. The figuration is this [plays the figured version of

the bass line], but that really amounts to a simple stepwise melodic progression [plays the underlying stepwise progression]. Little simple things like that allow him to think in terms of a line, think about where he's going to take it, and then the figuration will add rhythmic interest and will smooth out the melodic progression.

- 4.50** It's the same with the vocal parts in this cantata. So if you look at what the soprano is doing, it's this very ornamented line on the words, "Wir eilen," "we hurry." And of course, here again, he's attempting to illustrate that concept of hurrying and hastening with this very twisty, turny line [plays the ornamented soprano line]. Let's hear that again [plays the ornamented soprano line again].
- 4.51** And there again, that line is, if you boil it down, very similar to the bass [plays the underlying soprano line], but with lots of figuration added [plays the ornamented soprano line]. If you take those two parts and put them together and look at this resulting skeleton, you get something like this [plays the combined underlying lines].
- 4.52** It comes back to what we looked at in the Two and Three-Part Inventions and Sinfonias, where Bach is thinking in terms of simple ideas, the Invention, just a very simple line, in this case, a conjunct line [plays a simple conjunct line], and elaborating upon it in all the parts.
- 4.53** So I can't really tell you exactly how he wrote. We don't have sketches, as I mentioned. But if I had to guess, then I would imagine he would plot out a piece like this, first, in terms of, he would choose a key, and he would decide what the form scheme would look like and what the modulatory scheme would look like. In other words, "We're going to start in B $\flat$, then we're going to move towards G minor, and then maybe we'll have an episode in E $\flat$, then we'll go to F, and then maybe we'll go to D minor, and then we'll end up back in B $\flat$ at the end," something like that.
- 4.54** But it would be like if you're planning a trip, you would decide where you're going to stop along the way. You don't necessarily have all the details of the itinerary worked out, but you have a sense of where you're going and what the pit stops are going to be, something like that, an overall vision of how the tonal trajectory of the piece is going to evolve. Then he would have these simple melodic ideas and see how he could propagate them throughout the piece. That's as close as I can get to understanding how he might have been working.
- 4.55** I mentioned that Jesu, der du meine Seele is a chorale cantata based on a single hymn. As you would expect at the end of this piece, there's a four-part chorale where all the singers get up and sing together, and we hear the unvarnished, simple hymn tune in a straightforward presentation in a four-part setting. So I'm going to play for you at the keyboard the closing hymn, Jesu, der du meine Seele, and then I'll play for you the sung version so you can hear the difference [plays Bach's Jesu, der du meine Seele].
- 4.56** Let's hear what happens in the orchestra when all the voices and instruments play that together [recording plays].

- 4.57 I love how that one ends. The last words you hear at the end of this cantata in the final chorale are "in der Süßenewigkeit," which means "in sweet eternity," rather a nice way to end a piece, I would think. That's a preliminary vision of what goes on in many of these cantatas, but there are so many of them, and they're so interesting, and they're so varied in their approach that I can only encourage you to hear many of them, as many of them as you can. But that's one I particularly like. I'm happy to take a couple of questions if there are any points you'd like me to address.
- 4.58 To what extent do recordings today—I mean, they've been edited, mastered, and so on—reflect how Bach and his audiences would have heard these pieces of music at the time, both because of the church setting, I guess, and the acoustics there? And then also the preparedness of the performers, there's only a few days—I mean, they're professionals, but there's not much time to prepare. Do we know much about that?
- 4.59 We know that Bach was constantly complaining about the low quality of the performers that he had to deal with. Yeah, there are letters from him where he's moaning about how incompetent they are. But imagine what it would have been like to have Bach as your choral director and to have to deal with these 25-minute pieces, and they're difficult every week, and to have maybe two days to rehearse them, it can't have been easy.
- 4.60 As far as the performance circumstances, it's important to bear in mind that when you write a piece of music, you're writing not just abstract notes, but you're writing in terms of an acoustic. There's the sound, but the sound doesn't exist in a vacuum. It exists in an auditorium or in your living room, or in a cathedral, and you don't write the same way depending on what the venue is. If you're writing a piece that's going to be sung in Notre Dame Cathedral, where the resonance time of an impulse might be 30 seconds long, then you can't have this elaboration. You can't have fast rhythms. It just becomes mud. It becomes completely indistinguishable. So it conditions the style that you write in.
- 4.61 There are pieces by great Renaissance masters like Gabrieli, where the choruses are laid out in a stereophonic position throughout the church, where there might be multiple different groups of players all interacting with each other spatially, extraordinary spatial effects. That's something that got flattened out in a lot of the music of the 19th century, where composers were no longer explicitly thinking in terms of sounds propagating in space and the possibilities of a particular piece being played in a particular architectural environment.
- 4.62 Some composers were thinking about this. There was an American modernist composer called Charles Ives, who is one of my favorite musicians of all time. He wrote absolutely wild pieces where there would be instruments off stage in the wings, and there are all these fascinating effects of proximity and distance depending on where the players are laid out in the hall. He would have multiple different orchestras playing

at the same time at different speeds, crazy things like that. So some composers are aware of the possibilities of space and integrate it into their pieces.

- 4.63 But for the most part, despite Bach's intention to supply the Thomasschule and the Kirche with an endless supply of liturgical pieces, in reality, most of these pieces were only heard once. They weren't actually repeated. In fact, when Bach died, a rather sad thing happened, which was that the estate, which consisted not only of his instruments and his personal possessions, but all his manuscripts, was divided up amongst his children. We've lost at least a hundred cantatas as a result of this.
- 4.64 Some of the pieces were sold by his sons. Some of them have disappeared. Some of them, no doubt, were destroyed during the two world wars. Some may be in a library somewhere awaiting discovery, but so far, there have been almost no major discoveries of works of this sort. So a substantial portion, at least a third of the cantatas, are probably gone forever. These pieces would have existed in one copy. They wouldn't have been broadly distributed, and they weren't published. So if all you have is a manuscript and a set of parts, there's no safeguard at all. It's very easy to lose them.
- 4.65 The cantatas, interestingly enough, were not a priority for preservation when Bach died. You have to imagine this colossal mass of works that he'd left behind. In those days, it was very hard to preserve anything if it wasn't published. So the cantatas, which were perceived to be a fairly ephemeral, one-use type of compositions, were simply not given the same degree of care and caution that other major works of his were given.
- 4.66 The performers would have had maybe one or two days with these pieces, and the audiences would have heard them once. There's no reason to assume that the contemporary audiences in those sitting on these uncomfortable wooden pews in these cold, unheated cathedrals would have necessarily been having the time of their lives when they heard these 25-minute compositions and the sermon every Sunday. I suspect that they were listening, but probably not all that attentively.
- 4.67 Contemporary descriptions of what it was like being in those churches indicate that people would turn up late. They would be talking with their neighbors. They'd be drifting in and out of concentration. The circumstances for the performance of these pieces were not exactly ideal, which makes it all the more amazing to me that he poured as much effort into these works as he did. He had to have had unshakable faith that these works would eventually find a more permanent place in the repertoire.
- 4.68 But there again, that's his indomitable conscientiousness. Never any sense that, "Oh, this is just going to be done once, so we'll just phone it in." He didn't do that. Every compositional project, no matter how fleeting or ephemeral the occasion, was given extraordinary care. That's something that in my life as a musician, I also find quite inspiring.

- 4.69 I was wondering, how much does the timeless nature of Bach's works relate to the timelessness of what he's trying to portray as in the religious text? Those ideas are quite timeless as well.
- 4.70 Well, there's loads and loads of religious music that isn't timeless at all, that has aged very badly. Again, the vast bulk of music that was produced—he wasn't the first Kantor of the Thomasschule. He was number 25 or something, and all the people who occupied that position before him also had to write music.
- 4.71 Remember that Bach was part of a very extended family. You can go online and look at his family tree. It's insane. There are dozens and dozens and dozens of composers named Bach, of which he's one. And then it continued after him with his children and grandchildren and so on, and they all produced bodies of work. So there's lots and lots and lots of liturgical music that was written, and much of it is not a very good quality.
- 4.72 So the timelessness of it—well, he obviously took it very seriously. And even though the texts he was given to set were not always of the highest literary quality, nevertheless, he transcended them and created these incredible pieces. So I don't think there's a direct aesthetic link between the nature of what's being sung about, the timelessness of those themes, and the aesthetic quality of the work.
- 4.73 You can also get a sense of that by walking into any major European museum and looking at the painting collections. You're going to see a lot of paintings in the Louvre, for example, that are not because they're necessarily great paintings, but because they belong to this or that royal collection, and that's where they are. Some of the paintings are extraordinary and very powerful, and some are really mediocre. But they've been part of the collection for 300 years, and there they are. So timeless concepts and ideas don't automatically result in timeless artworks.
- 4.74 I'm curious about the text of the cantatas. You mentioned they were written by second-rate poets. Would they write on commission from Bach, or was it more a case if he was choosing from a pre-existing anthology of poems and then putting music to those words?
- 4.75 It was both. Bach certainly had a library of texts that he kept around that he would use, he would dip into when he was looking for text to set. So for example, he had collections of hymns, some of which were from the time of Luther and some of which were more recent. But in this case, the hymn that he set was from 1641. So by that point, it was an 80-year-old text.
- 4.76 And so he would get a librettist—and they weren't all mediocre. Some of them were of higher quality, and once you become familiar with the texts and the cycles of cantatas, you can see the difference in quality. There was one author that he worked with quite a lot called Piccander, which is a nom de plume. It's an artist's name. And Piccander wrote the libretto for the St. Matthew Passion, which is a great text. So they're not all homogeneously bad, but some of them are pretty clunky.

4.77 So Bach worked with a mixture of existing poems that were already published that belonged to his library and then contemporary poets that he either worked with or knew personally.

Lecture 5: St. Matthew Passion

At a glance — A close look at the St. Matthew Passion, one of Bach's supreme achievements, and its dramatic and devotional scale.

- 5.1 We've been taking a chronological tour through Bach's work, through the evolution of his style, and also through the multiple facets of his compositional project, which was vast in its scale and its ambition. Yesterday, we looked at one of Bach's approximately 300 cantatas, *Jesu, der du meine Seele*, which dates from 1724. Today, we're going to move a little bit forward, three years later, and look at one of his most imposing monuments, the St. Matthew Passion, which was composed in 1726, and looks to have been premiered in 1729, two years later.
- 5.2 This work really stands apart from everything else Bach ever composed. It's an immense piece, a towering masterwork that challenges audiences even today. It may be the most ambitious project he ever undertook, possibly rivaling the B Minor Mass, which is one of his late masterpieces written towards the end of his life. The St. Matthew Passion is really unparalleled for the breadth of its ambition, its scope, its dramatic intensity and its sheer duration. This is a piece of music that lasts for over two and a half hours in a performance.
- 5.3 Bach spared absolutely no effort in going as far as he possibly could with this project. In fact, when you look at the original manuscript for the St. Matthew Passion, it's obvious that this piece stood apart for him. He took extra care with this manuscript. The text of the score is written in multiple different colors. It's beautifully calligraphed, and he took great pains to ensure that it was properly preserved.
- 5.4 In addition to the vast scope of this project, retelling the passion story, Bach decided to do something extraordinarily extravagant. The piece is written for a double orchestra, two complete sets of instruments, situated antiphonally in the hall, which is to say that there are effects of stereophonic motion between left and right. So two complete sets of instruments, each of which has their own continuo, which is to say the figured bass line that we mentioned yesterday—so a bass instrument and a polyphonic instrument to harmonize the line—a double choir and an additional soprano choir. All of these forces produce this extraordinarily intense maelstrom of sound.
- 5.5 Bach makes sure that throughout the two and a half hours of this work, the texture is continuously changing. So there are passages of great intimacy, where we mainly hear a tenor recitative, accompanied with the continuo, nothing else. There are arias and ariosos, where a solo voice stands up with an instrumental accompaniment.
- 5.6 Of course, there are choruses that feature the entire forces of the ensemble and everything in between so that this two and a half hour span of music is not articulated in one uninterrupted block, but rather in dozens and dozens of individual numbers, each of which has markedly different forces over the course of the piece.

- 5.7** Sometimes this double orchestra and double chorus are heard simultaneously. Sometimes you hear only the one on the left side or the other on the right side. Sometimes they mingle. It's continuously changing. This is, by any measure, an extraordinarily ambitious project. Bach had written passion music previously. The St. John Passion was a piece that contrasts notably with the Matthew Passion in that it has a very strong sense of narrative impulse, always moving forward, and the action progresses at something of a rapid clip. That's not the case here at all.
- 5.8** One of the things that makes the Matthew Passion both an exhilarating experience and also one that can be challenging for audiences is the fact that Bach divides up the story into the narration of Matthew's words, delivered by a figure called the Evangelist, which is a tenor solo. The narrative aspects of the piece are primarily given in these recitatives.
- 5.9** But rather than a straightforward presentation of the narrative of the story, Bach continuously interrupts the flow to give us sections of the piece that are really more contemplative or interpretive in nature, where he has a soloist or a chorus comment upon what's just happened, reflect on it. So as this extraordinary narrative is unfolding, you have this strange stop-and-start rhythm where we have to think about it. We have to contemplate what's happening. We have to reflect on it. He'll also personalize the emotions of the story through these arias and ariosos in which a soloist incarnates the emotions, the extreme emotions of this story.
- 5.10** So it's a piece that's quite demanding on the listener's concentration in certain respects. I've certainly heard live performances of this piece—where I live in Strasbourg, the entire thing is played every year, and it's a major cultural event. You enter into this meditative tapestry. I don't know how else to describe it. It's not operatic. It's not a straightforward narrative. You have these shards of narrative and then a whole interpretive and reflective framework around it, and you keep alternating between the two. As the two and a half hours wear on, you enter into this strange space that seems to be outside of time, while still having, of course, the outline of the Passion story.
- 5.11** By this point, in 1727, Bach had more or less completed his cantata cycles; he's written hundreds of them. He knows this genre inside and out. He's experimented, he's tried all sorts of things, and he's well into his 40s by this point. The St. Matthew Passion is an attempt to summarize everything he's learned and also to create extraordinary dramatic effects.
- 5.12** Some of the numbers included in this piece are among the most famous pieces that Bach ever wrote. Some of his most beautiful, intense, passionate, and moving music. In fact, the St. Matthew Passion, for all of its difficulties and peculiarities as a piece, was really the piece that led to Bach's reputation coming back to the forefront of the musical world.

- 5.13** There's an interesting story about that. A hundred years after its premiere—in 1829 a young German musician called Felix Mendelssohn became interested in this piece. He got a hold of a copy of the score, was fascinated, and decided that this should be brought to public attention again. Remember that at that point, in 1829, Beethoven was already dead. The music world had moved on. The style of Bach was considered to be completely old-fashioned and anachronistic. It was a different era.
- 5.14** These days, everything exists all the time in a permanent archival state. Nothing is lost. We have access to the world's music and the music of all the different epochs of human history at our fingertips. There's this strange, in the classical music world, this strange situation, a bifurcation, if you like, between the repertoire, the museum culture of preserved past masterpieces, and then the production of new artists.
- 5.15** It might be a bit much to say there's a conflictual relationship between those two things, but there's a strange incompatibility in a way where the past is constantly threatening to devour the present. We have this accumulated mass of great works from the past, and it can make it quite difficult for living artists to produce their work. You're not just competing, in other words, with your colleagues anymore, but with the totality of everything that's been produced in the past.
- 5.16** But in 1829, that wasn't the case. If you went to hear a concert, in all likelihood, you'd be hearing a new piece. This concept of the repertoire of a body of past works that have been filtered out through criticism, through audience selection, through the esteem of other artists, was not such a prominent idea. There is nothing to guarantee that the work of a composer like Bach would exist in perpetuity and would be heard by people.
- 5.17** So Mendelssohn looks at this score, and of course, by then, the Baroque era is well in the past, and nobody plays continuo parts anymore. Nobody plays the harpsichord. The orchestral instruments of Bach's day were no longer played—the old oboes and oboes d'amore and the old boxwood flutes and so on.
- 5.18** So the piece can't be played exactly as Bach wrote it because the concept of period instruments didn't exist at that time, the idea that you would conduct musicological research into the music of the past, try to determine how it would have been played, under what circumstances, in what type of venue, and on what instruments. There's a whole army of scholars today that work on these problems. Then there are luthiers, instrument builders who specialize in recreating old instruments.
- 5.19** But again, in 1829, that wasn't the case. You had to make do with what you had. So Mendelssohn made an arrangement, a version of the score that could be performed with the instruments that they had, and performed the piece: the St. Matthew Passion. Remember, at the time, Bach simply wasn't a household name at all. The re-performance of this piece for the first time in 100 years made a colossal impression.

- 5.20** It's actually thanks to Mendelssohn, a composer, that not only did the work survive, but this initiated a whole musicological industry of scholars attempting to piece together Bach's life and works, trying to locate as many manuscripts as possible, keep them in a state of preservation, and edit and publish all of these scores. Almost nothing was published in Bach's lifetime. If you wanted to play one of these pieces, you had to actually have the original manuscript or copy it out.
- 5.21** So the Bach Gesellschaft, the Bach Society, was founded with the express aim of rounding up all of these scores, attempting to make authoritative editions and make them widely available. But that happened on the heels of this immensely impressive performance. That could very easily not have happened. It's hard to overstate just how fragile and contingent these things are, how it routinely happens that works are forgotten for 200 years and, through some accident, are revised. That was the case with the St. Matthew Passion.
- 5.22** Today, if you had to choose a work that stands at the summit of Bach's achievement, many people would select this piece. Many commentators would even go farther than that and suggest that this is one of the absolute high points of the Western art music tradition. In this respect, I think this piece stands shoulder to shoulder with the later B Minor Mass, which is another work that is often cited as being one of his greatest—a complete, full Catholic Mass, nearly two hours in duration. But nevertheless, there is something about the St. Matthew Passion that keeps drawing us back to it.
- 5.23** So I mentioned the instrumental substance of this piece, the fact that we have a double orchestra, a double chorus, and what's called a ripieno chorus, or a smaller additional chorus of sopranos—it would have been boy sopranos at the time. And from this group of singers are drawn various soloists who represent the different characters in the story: Jesus, Judas, and so on. There's one soloist in particular, whom I mentioned earlier, the Evangelist, who is present throughout the entire piece, narrating the action as it unfolds.
- 5.24** It's a challenging piece to come to terms with. You have to distance yourself from the expectation that we're going to have a continuous flow of narrative. You have to get into this rhythm of a little bit of story, an extended commentary on the story, and also choruses and chorales, communal moments, where everybody gets up and sings together. Through this continuous change of textures in the piece, we have all of these multiple different perspectives running along in parallel to each other, moments where the emotions of the story are made concrete by a single voice, moments where the collective representing perhaps all of humanity stands up and sings together.
- 5.25** I'm going to play you a few excerpts from this piece. It's complicated to choose excerpts from such a gigantic work, but I'm going to play three that give you at least some sense of what it's like. We're going to start with the opening chorus. The opening chorus is a mind-boggling composition. It lasts about five minutes, sometimes a bit more, depending on the tempo.

- 5.26** It's not just a question of the emotional power of this movement, which was necessary, because when you're beginning a two-and-a-half-hour-long piece, telling nothing less than the story of the trial, the crucifixion, the death, the burial of Christ, you can't have some little one-minute thing. It has to be grand. You have to signal to your listener immediately that this is the story of stories. This is something immense. This is an epic. The opening chorus fills that function admirably well, but it also has elements that I think are astonishingly modern.
- 5.27** I mentioned that there's this antiphonal aspect where you have different sound sources in opposite ends of the cathedral or the hall in which the piece is being played. That in itself was nothing new. We mentioned earlier that in the music of Gabrieli, for example, a Renaissance composer, that he would explicitly make use of the space of the architecture in which a piece was being played to position singers in different parts of the cathedral and obtain all of these spatial effects. In the 20th century, that again became an idea that was preoccupying composers.
- 5.28** There's a quite famous piece that was premiered in the late-1950s called Gruppen or Groups by Karlheinz Stockhausen, a German composer born in the 1920s, 1928, I believe, where there are three separate orchestras placed on three separate plateaus in different spatial locations of a concert hall, each of which has its own conductor, each of them playing at different speeds and changing speeds relative to each other. It's a pretty phenomenal piece of insane ambition.
- 5.29** When I think about things like that, that were done in the mid-20th century, I think, "Yeah, that's incredible, the technical accomplishment, the ambition of it, the scale of it." But then I relate it back to things like this, with this dizzying maelstrom of sound.
- 5.30** There are effects in this opening chorus that I don't even know how to begin to describe. There are these bass lines, for example, that seem to be spiraling upwards forever, just keep going. There's a rising motion that's very characteristic in this piece. It gives it a very strange quality because, on the one hand, he needs to establish the gravity of the subject.
- 5.31** You have this throbbing bass line in 12/8, which is an unusual meter: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12. It's this slow, almost funereal procession with these repeated notes in the bass, which you're going to hear in a second. It establishes immediately this vastness of scope, this grandiosity. But then you also have this continuous spiraling, upward motion. As with the cantatas, the St. Matthew Passion is full of musical figuralisms, full of devices, musical devices that are intended to convey the emotions of the story, not only through words, but also through sounds.
- 5.32** So this opening chorus is titled, Kommt, ihr Töchter, helft mir klagen, which translates to, Come, My Daughters, Help Me Lament. I'll read you the text so you can get a sense of what's going on. "Come ye daughters, help me lament. Behold, whom? The bridegroom. Behold him. How? Like a lamb. Behold, what? Behold his patience. Behold,

where? Behold our guilt. Behold him out of love and graciousness. Himself carrying the wood of the cross. Oh guiltless lamb of God, slaughtered on the stem of the cross, always found patient, although thou was despised. All sin hast thou borne, else we must have despaired. Have mercy on us, O Jesus."

- 5.33** Let's listen to the opening chorus of the St. Matthew Passion. As you listen, try to focus on not just the continuous unfolding of this immense polyphonic texture, but also the antiphonal effects, where he has one chorus make a statement like, "Behold," and then, the other chorus, on the other end, answers. Constant question and response type of dynamic in this piece. So here's the opening chorus of the St. Matthew Passion [recording plays].
- 5.34** So from there, Bach continues with a sequence of recitatives, creating this gigantic contrast between the full forces of the double orchestra, the double chorus, the spatial effects and everything, and then this very intimate scale, where we just had a spotlight on the story.
- 5.35** Remember that when Bach applied for that position in Leipzig, he was the third choice after Telemann and Graupner. It's funny to contemplate, maybe not funny for him, but you can go and listen to the music of Graupner. You can find it interesting to compare the two.
- 5.36** One of the things that for me as a composer is so fascinating is this constant telescoping between extremely different scales, between this massive, gigantic polyphony with its complicated harmonic trajectory, and then these moments of intimacy and simplicity, where he has these simple chorales. I'm going to play you one of them, which actually appears twice in the St. Matthew Passion, and it's called *Herzliebster Jesu, was hast du verbrochen*.
- 5.37** I'm going to read you the text. This is a collective meditation on the unjust nature of the punishment of Christ. The words are: "*Herzliebster Jesu, was hast du verbrochen, dass man ein solch hart Urteil hat gesprochen? Was ist die Schuld, in was für Missetaten bist du geraten? Herzliebster Jesu.*" That translates to: "O dearest Jesus, what law hast thou broken? That's such a hard sentence has been spoken. Of what great crime hast thou to make confession?"
- 5.38** I'll play you the melody of this chorale, and then we'll listen to Bach's harmonization of it [plays the melody of *Herzliebster Jesu, was hast du verbrochen*]. So there we are, again, zooming out from this massively complex polyphony to this moment of simple humility and directness from a six- or seven-minute-long opening chorus to this tiny little *joule* that's less than a minute long.
- 5.39** I mentioned that in addition to these massive choruses, chorales, and the recitatives, in which the story gradually unfolds, we also have moments of emotional clarity in which a soloist stands up out of the chorus and sings an aria or an *arioso*. One of the most famous arias, if not the most famous aria that Bach ever wrote, is from the Matthew

Passion. It's a piece for alto voice and violin plus instrumental accompaniment called *Erbarne dich*, Have mercy.

- 5.40** This comes in the second half of the Matthew Passion, and what it is really is a reflection on Peter, who has denied Christ three times, runs outside, and weeps bitterly. The text is very short, but quite powerful: "*Erbarne dich, mein Gott, um meiner Zähren willen; schaue hier, Herz und Auge weint vor dir bitterlich.*" "Have mercy, my God, for the sake of my tears. Look hither. Heart and eyes weep before thee bitterly."
- 5.41** As you would expect with Bach, he's always concerned not just with abstract musical ideas, of course, but with creating the most intense possible effect or affect through his musical devices. He always uses these figuralisms to try to make the action, the emotion, or the story palpable, even if you don't understand the words. In the case of *Erbarne dich*, Have Mercy, the tears, the weeping of Peter is symbolized through this solo violin that runs throughout the whole piece.
- 5.42** There's a little motif, a melody played on the violin that recurs again and again throughout the piece. You might initially think this is a *ritornello*-type form, which is derived from Italian concertos, where you alternate a recurring theme with short episodes in between. It has some commonality with that, except that in this piece, you don't have contrasting sections in the way that you would in a true *ritornello*. There's an *obligato* line on the violin that keeps recurring, but instead of having sectional contrasts, the contrast that you might have in the piece are absorbed into this polyphonic texture with multiple lines going on at the same time.
- 5.43** One of the most remarkable aspects of this piece, I think, is the way this solo violin line, with its gorgeous melody and ornamentation, gives us a palpable sense of this bitter weeping. In a way, he's attempting to show us also that there are emotions, there are experiences so intense that they cannot be communicated with words.
- 5.44** So when the alto voice finally starts singing after the violin introduction, it starts with the incipit, the beginning of the melodic line. We expect, as would normally happen, that the voice is going to take over the melody and restate it in the same way. Instead, what happens is that after a few notes, the alto voice doesn't get stuck exactly, but wanders off in another direction and becomes more of a counterpoint to the violin line, which continues.
- 5.45** It's as though Bach is telling us that the only way to state this is through sound. The words are not going to be enough. And throughout the piece, of course, the voice is a prominent element, but it's as though the truest eloquence is reserved for the violin.
- 5.46** So we're going to listen to *Erbarne dich*, and I'm going to ask that you focus on the figurations, the figuralisms, rather, in the violin part, how we have this palpable sense of this emotional outburst, and on the interplay, the subtle relationship between the

voice and the violin. Here is Erbarme dich from the second half of the St. Matthew Passion [recording plays].

- 5.47** One of the things I take away from this piece as an artist is a sense of how ambitious it's possible to be. You can think about what art can do and what function it might have. And Bach was clearly animated by not only an intense feeling of devotion and religiosity, but a sense of mission. It was very, very strong. That obviously pushed him to do things that I don't know if you can do them without that. I'm not sure. Maybe not.
- 5.48** In a town called Colmar, which lies about maybe 35 minutes outside of Strasbourg, where I live in Alsace, there's a very famous artwork called the Retable d'Issenheim, the Altarpiece of Isenheim, which is a triptych painted by Matthias Grünewald, a Renaissance painter. It depicts the crucifixion, but in these wild, psychedelic pastel colors. It's a very, very strange painting, exceptionally strange, all these mannerist, twisted, contorted figures.
- 5.49** You look at things like that, and they're so striking. They're so shocking even today. The effect is absolutely overpowering. That approach to artwork, where there's this powerful, symbolic dimension, where the technical and expressive means are used to create a literally overpowering effect, that's something that always deeply impresses me. It's hard to scale down your ambition after that, once you hear things like this.
- 5.50** So that's a brief introduction to the St. Matthew Passion. But really, I have to say that the way to hear this piece is in a performance. I personally find it difficult to listen to very long pieces on a CD or on Spotify. It doesn't replicate the experience. For example, there are operas by Wagner that are four hours long. I can't for the life of me understand how, in the old days, people would get these huge box sets of LPs and sit there and listen to one side after another for four hours. That's completely bizarre to me, but people do do that.
- 5.51** But really, the experience of this piece is a collective experience where you sit in the space, and you hear the sound propagating in this acoustic environment, in this architectural space. There's no substitute for that. There's no substitute for the sensation of time very slowly unfolding with this strange meditative pace that the piece creates.
- 5.52** So if you do nothing else musically, I do highly recommend that you try to get yourself to a live performance of the St. Matthew Passion. I believe this is one of the high points of Western civilization, of Western culture. It's one of the great artworks, and it's something that I think everybody should experience once. We've got a few minutes left for questions.
- 5.53** With St. Matthew's Passion, you've mentioned the symbolism, the emotional depth, the ambition. Do you think that's linked to the religious influence, or is it possible in a secular composer or society?

- 5.54** It's a complex question because if you look at Bach's secular works, you do see a similar level of ambition. We're going to look at some of those pieces later on. But I wonder if you can so easily separate the sacred and the secular, actually. I wonder if the apparently secular works aren't animated by the same sense of piousness, the same sense of devotion, the same sense of being called upon to marshal the fullest resources of his skill and his effort at all times.
- 5.55** The fact that it never seems to wane, the fact that he got up day after day, year after year, got to his work desk and produced these pieces, indicates to me that he must have applied that sense to the secular pieces as well. But clearly, it's like a motor. It's like an engine that not only Bach, but that artists throughout the centuries have drawn upon to push them, to push them farther, always.
- 5.56** The intense love and care that Bach put into this manuscript is also something that has to be seen to be fully appreciated. You can go online and find complete facsimiles, complete reproductions of Bach's manuscripts. This one in particular is simply spectacular. It's an amazing bit of calligraphy.
- 5.57** So much so, actually, Bach revised the piece, about 10 years later, in 1737, he made a revised version of it, expanded it, added pieces to it. He was clearly very focused on this piece. It was obviously, for him, one of his crowning achievements. It's impossible to imagine it, I think, without that religious underpinning. I don't think anything like it could have been produced otherwise. I don't believe so.
- 5.58** If you were to go to a live performance for the first time, what level of preparation would you recommend putting in to experience it in a maximally meaningful way?
- 5.59** This has often been a dilemma for me as well. When you go to hear an opera, for example, the odds are it won't be in English because there are very few operas in English. It'll be an Italian, German, or maybe French. And if you don't speak the language, then you have to deal with the fact that there's the action unfolding on the stage—there's the costumes, there's the acting—there's the musical dimension to it, there's the narrative, and then it's all unfolding in a foreign language. Especially if you have duos and trios and so on, where you have multiple characters speaking at the same time and interacting with each other, then it quickly becomes an unbelievable sensory overload where you can't really keep up with it.
- 5.60** There are people who are maybe more studious than me, who go and get the libretto and read the whole thing beforehand so that they understand the story and the characters. You can also read the titles, which are usually projected in the hall, and so you can read a translation. But then at that point, for me, it becomes impossible. I can't listen to the music, watch the stage, follow the story, and read a translation all at the same time. I can't do that.
- 5.61** So how do you interact with an artwork like that? It's tricky because a piece like the St. Matthew Passion, again, its premiere in 1729, would have been done in a context

where everybody spoke the language, knew the story, and where you were in a familiar setting. We simply don't have that context anymore. It's inherently challenging.

5.62 It's tempting to say you can dive in and experience the force of this piece without necessarily having to go and do two weeks of homework beforehand. Then what you can do—I mean, what happens in my case, much of the time, is I just go in cold, basically. If I'm really intrigued by the piece, then I go afterwards and read about it, and I might listen to it again. I might try to attend another performance.

5.63 But the idea that you could, in our present context, 200 years removed from when this
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Lecture 6: Goldberg Variations

At a glance — The Goldberg Variations, prompted by a question on the role of religious thought in Bach, and the variation form as a vehicle for invention.

- 6.1 We had a question last time that was quite interesting about the importance of religious thinking in Bach and whether or not that would have influenced the scale of his ambitions. This next work that we're going to look at, I think, is an indication that even in a completely secular context, Bach was quite capable of pushing things to almost an absurd degree. There's an amazing story about this piece that is almost certainly apocryphal, but I'm going to tell it anyway.
- 6.2 We're getting now into Bach's final decade. As the years in Leipzig wore on, Bach started to have fewer and fewer obligations. He started to be relieved of some of his duties. He was no longer having to spend his days rehearsing the apparently incompetent musicians that were in his charge, and he starts having fewer diplomatic spats with the powers that be in Leipzig. By this point, he's a well-established and very eminent figure, and he's given a little bit more freedom, which I suppose is fair enough at that stage.
- 6.3 In the last 10 years of his life, in the 1740s, there's a major shift that starts to happen in his production. He's no longer required to produce these mammoth cycles of cantatas. He's really fulfilled a lot of his compositional ambitions by this point. So in the last 10 years of his life, his output starts to not exactly slow down, but to become concentrated on a smaller number of works. He throws himself into these projects with extraordinary intensity.
- 6.4 So rather than the furious pace of production that characterized the 1720s and 1730s, by the 1740s, he's working on these, what I called earlier, these encyclopedic projects that would allow him to go very deep into a very specific and narrow question. He produces six major works in the last 10 years, as opposed to the hundreds and hundreds that he produced in the preceding decades. He really focuses his efforts on them.
- 6.5 You remember that one of the ongoing projects that Bach had was the series of Klavierübung pieces, keyboard exercises. In 1741, Bach embarks on the fourth and final installment in that series, which is the Goldberg Variations. The Goldberg Variations is a title you may have heard before. It's one of the most famous pieces of Bach, which is interesting in and of itself because it's not a piece that necessarily lends itself automatically to that level of popularity. It's a very strange piece in some respects.
- 6.6 But nevertheless, this work was written in 1741, and it consists of an aria, a melodic piece for keyboard, followed by no fewer than thirty variations. This, again, this piece has an encyclopedic scope in that if you look at all thirty variations, what you see is a

compendium of every conceivable Baroque dance style and every conceivable national style as well. There are French overtures in this piece. There are gavottes.

- 6.7 There are canonic variations also, where, amazingly, Bach writes a series of canons in the Goldberg Variations, each of which is at a different interval and spaced out every third variation in the cycle. There's this characteristically elaborate and thorough architecture to the work. Variation 3 is a canon at the unison. Variation 6 is a canon at the second. Variation 9 is a canon at the third. Variation 12 is a canon at the fourth and so on. It goes all the way through, all the way up to the canon at the ninth towards the end of the piece. It's a piece of absolutely dizzying ambition on every level.
- 6.8 The story behind the creation of this piece is funny. It's almost certainly not accurate. It was reported by a biographer of Bach, who never met him personally, but had some dealings with Bach's sons, notably Carl Philipp Emanuel Bach, who himself was a highly successful composer, writing in a completely different style from his father—we'll get into that a little bit more later.
- 6.9 In the book, this biography of Bach, that was written after Bach's death, he relates the following story: there was a Russian ambassador to Leipzig who supposedly suffered from insomnia and wanted a long keyboard piece that could be played to him at night to help him fall asleep. Bach was tasked with writing this series of pieces, which were to be played by a harpsichordist named Goldberg, hence the name Goldberg Variations.
- 6.10 The story doesn't quite make a lot of sense, and it's also made even more doubtful by the fact that Goldberg, at the time, was 14 years old when Bach was writing this, which, when you consider the extreme difficulty of this piece, it's not really plausible that he would have been able to play them. In fact, it's hard to imagine how any contemporary keyboardist at the time could have managed some of these pieces.
- 6.11 So one of the things that distinguishes the Goldberg Variations, but also led to these pieces not being widely disseminated at all, is the fact that Bach pushes keyboard technique to its absolute limits. These are incredibly virtuosic pieces, some more than others. But the most difficult pieces in the series are phenomenally difficult to play.
- 6.12 And not only that, they require what's called a double-manual harpsichord. If you've ever seen a church organ, you'll have noticed that organs typically have multiple different keyboards, one on top of the other, and they're recessed so that you can play on different keyboards at the same time. The reason for that is so that you can have different what are called stops. You can have different registrations so that one keyboard sounds maybe more like a flute, and the other keyboard sounds more like a brass instrument or like a reed instrument. You can have these mixtures of different timbers. You can change the registration from piece to piece. You can have one timbre in the left hand and another in the right hand.
- 6.13 You can also have organum-type effects where you have one keyboard doubled at a fixed interval by another set of pipes, doubled at the octave or at the twelfth or at the

double octave and so on. So you can have these really beautiful, rich mixtures of sounds. That's why you have sometimes hundreds and hundreds of pipes in these instruments, because there's not just one pipe per note, but there are multiple different keyboards and multiple different stops, entire sets of sounds that each require their own set of pipes.

- 6.14** The Goldberg Variations requires a special keyboard, a special harpsichord, that is a double manual instrument. That means there's one keyboard and another on top. The reason that's important is that the piece contains multiple passages where the hands have to cross like this, and you have to do all of these acrobatic runs and arpeggios. That's very, very difficult. On top of that, if you try to play it on a modern piano, it's possible to do, and people do do that, but it's extremely challenging because your hands are actually like this a lot of the time. You have to find a way, ergonomically, to make it all work; it's much easier on a double manual harpsichord.
- 6.15** So this piece languished in obscurity for a long time, simply because it's so unbelievably hard to play, and was rediscovered and became popular again in the 20th century. One of the recordings that caused that renaissance to happen was the famous Glenn Gould recording of the Goldberg Variations. Here we come back to Glenn Gould again—Glenn Gould was a central figure for me in my life in terms of becoming interested in Bach and also coming to terms with these great pieces.
- 6.16** In 1955, Glenn Gould, who was 22 or 23 years old, very, very young, brilliant pianist, obtains a recording contract with Columbia Records and is asked, "What would you like to record first for your debut album?" And he says, "I want to do the Goldberg Variations." At that time in the mid '50s, 1955, that was not how you started your career. You don't start your career with this gigantic, fairly recondite collection of Baroque pieces that was not particularly popular. It had this reputation as being an austere, cerebral piece.
- 6.17** Pianists' stock-in-trade is the 19th-century repertoire. Pianists live off of Chopin and Liszt and Schumann and all of these great romantic pieces that people love to hear and that constitute the basis of a lot of pianists' technique. The Goldberg Variations—it was hard to make a commercial case for that piece, especially from a young pianist starting their career.
- 6.18** So Columbia Records hesitated, resisted, and said, "Well, wouldn't you rather do the Chopin nocturnes instead?" No, he really wanted to do the Goldbergs. He made a recording of the Goldberg Variations in his early 20s, and it was spectacular. Absolutely spectacular. He played these pieces with almost recklessness at top speed with incredible rhythmic verve and energy in a way that immediately spoke to audiences and caused these pieces, the different pieces of the Goldberg Variations, to become a sensation.

- 6.19** Nobody had ever played Bach that way before. There were pianists and harpsichordists, keyboardists who played, of course, the works of Bach. But if you listen to earlier recordings than that, recordings from the 1940s or the early '50s, very often they're played in this dry way. Not terribly exciting—accurate—but you could see why Columbia Records might have hesitated.
- 6.20** Gould played them with this abandon that was immensely appealing. The virtuosic variations, he absolutely threw caution to the wind and played them at insane speed. Gould had this phenomenal technique that was arguably more advanced than anybody else at the time. He could do these things and play them absolutely accurately at speeds that bordered on the ridiculous.
- 6.21** That made his career. Not only did it make his career, but it also, in a sense, threw the Goldberg Variations into the spotlight. Now, it's a piece that is very often played, very often recorded, by pianists. It's an acid test of: Can you contend with this piece? There are some very famous recordings of it now, including the jazz pianist, Keith Jarrett, who recorded not only this, but also the Well-Tempered Clavier and other works of Bach. Now it's an absolute staple of the repertoire. But it took a good 200 years for that to happen.
- 6.22** So it's a popular piece. It's a very appealing one in many respects, but it's also got its peculiarities. One of them is that the Goldberg Variations, as the title suggests, is a set of variations, and perhaps we'd better stop there and explain what that actually means. The theme in variations, variation form, by that point was a well-established genre in which a composer would take a melody and compose a series of variations on that melody, where each one has a different figuration, each one has a different texture, or maybe a different style.
- 6.23** There are many examples of variation form. You can look at pieces like the La Folia Sonata of Arcangelo Corelli, which is one of the first really popular Baroque pieces. It was a piece that was actually published and widely available and disseminated and could be purchased, and all of that, which at that time—I believe this is the late 1600s—was not by any means common. Most pieces were not published. Most pieces did not have any commercial success or even commercial viability at all. That was a set of variations, the La Folia Violin Sonata of Corelli.
- 6.24** But the difference, though, is that the Corelli piece is basically based upon a traditional melody. In the case of the Goldberg Variations, what he does is the variations are not really anything to do with the melody of the aria. Instead, what he does is he takes the ground bass, the bass line of the aria, and to a degree, the associated harmonic progression, and uses that as the basis for the variation.
- 6.25** When you hear the aria, and then you move to Variation 1, Variation 2, and so on, you're not going to hear any resemblance whatsoever between the melody of the aria and what happens in the successive variations, but you will hear the same bass

progression over and over and over again. That connects this piece with earlier forms like the chaconne or the passacaglia, both of which fundamentally have to do with a repeating bass line that loops around many times. Then on top of that, you have a series of different figurations and textures on top of it.

- 6.26** The other thing that typically happens in a classical theme variations is that you start with a very simple, clean, minimalist presentation of the theme. It has to be simple and clear. Usually, the melody is pretty straightforward as well. Then each successive variation will get a little bit more elaborate. Often by the end, you have some virtuoso showpiece. There's many, many examples of this.
- 6.27** There's a set of variations by Mozart on *Ah, vous dirai-je, maman*, which is a traditional French song. We know it as *Twinkle, Twinkle, Little Star*. It's a very cute little piece. There are also many Baroque sets of variations by composers like Rameau. There's a piece called the *Gavotte* with six variations that follow it, in all cases, the progressive elaboration of the texture. But Bach doesn't do that. He bases the whole thing on a ground bass, not on a theme. On top of that, he scatters these incredibly difficult variations throughout the whole work. There's no particular sense of progression as you move through the thirty variations.
- 6.28** So this whole story about this ambassador suffering from insomnia and all of that is probably an elaboration, probably. There might be some basis, in fact, but you might wonder, nevertheless, "Why did Bach do this? Why did he write thirty variations? Why does there have to be so many?" Again, I think it's simply a matter of his desire to always push a project to its ultimate conclusion and to leave no stone unturned. This is, again, the encyclopedic dimension of Bach.
- 6.29** This is where, with the *Goldberg Variations*, we come to a challenge for audiences, which is that thirty pieces for keyboard is a lot. On top of that, they're all in the same key of G major or G minor. That's a substantial amount of music based on the same bass line. Some commentators feel that the piece, in a certain sense, doesn't quite work because the constant G majorness of these pieces, one after the other, can become monotonous if you listen to them one after the other.
- 6.30** I'm a little bit on the fence as far as that goes. I love the *Goldberg Variations*. It's a piece that I've been listening to for thirty years now. I never get tired of it. I suppose whether you enjoy it or not depends on your tolerance for a very long stretch of music without much in the way of modulation. But I think that what Bach does—this is so interesting—if you can call upon your faculty of invention to create such a wide variety of effect, and texture and style, that the repeating elements in the piece, the ground bass, just sink into the background, you don't even pay attention to them anymore because you're absorbed by the continuous unfolding of the highest level of musical invention. That's hard to do.

- 6.31** I said in the composition course that a lot of what composition is is disguised repetition. It's the same with writing songs. If you were to take an average pop song and boil it down and look at what's actually going on structurally, you'd find that it contains maybe 45 seconds of material at best, because a lot of things are repeated: the chorus is repeated, the verses are repeated—you might have a chorus, a verse, a middle eight, maybe an introduction, and boil it down. It might really just be a few seconds of music; the rest is all repetition. You don't necessarily notice the repetition because the texture and the arrangement and the vocal performance, all of those things change and are dynamic. But from a structural point of view, there's probably not that much material in it.
- 6.32** That's the case here with the Goldberg Variations to a degree. The piece consists of a ground bass, which is 32 bars long, played over and over and over and over again in many different forms. What I'm going to do is I'm going to play you the ground bass at the keyboard, and then I'm going to play the aria, and then we're going to see what Bach does with this. Here's the ground bass of the Goldberg Variations [plays the ground bass of Bach's Goldberg Variations]. That's it. He writes thirty pieces and an aria based on that bass progression.
- 6.33** At the start of my composition course, I pointed out how a composer can take a line like that and turn it into something because it's not really the material in a sense that matters. It's how you handle the material and what you do with it. Bach might not even have written this bass line himself. It might not even be original. It doesn't matter in a sense.
- 6.34** I'm going to play you the opening aria of the Goldberg Variations. What you'll see is that this is a richly ornamented melody, and quite a beautiful one, I think [plays the opening aria of the Goldberg Variations]. It's a beautiful piece, but a lot of what's going on there is ornamentation. If you actually look at what the melody is actually doing, it's very simple, and some people feel it's not even the most interesting piece in and of itself.
- 6.35** The fact that he chose this as the starting point for a cycle of thirty variations is in itself interesting because it's pretty, but there's nothing particularly remarkable or spectacular about this tune. I want to just play the opening without the ornaments, and you'll see it's just a very simple little skeleton [plays the unornamented aria]. There's nothing particularly memorable, necessarily.
- 6.36** Although I have to say, when I first got the score and started working on some of these variations—I certainly can't play all of them; most of these are way beyond my limited technique—but I could play the aria when I was a teenager, and I was struck by one thing, which is that even in this supposedly straightforward melody and accompaniment texture, this piece is not particularly contrapuntal, but it consists of a clear division between the melody in the right hand and these accompanimental figures in the left hand.

- 6.37** Even in this piece, Bach does something quite interesting, which is where he takes this little bass line and occasionally adds a third voice that grows out of it. The way he does that is by holding some notes and then continuing to move from one note to another with the other part of the left hand. So you get these little effects of resonance, of a harmony emerging out of this linear passage in the left hand. I was really struck by this. I'm going to play an example so you can hear what I mean, but this idea of a latent polyphony growing out of what is actually a continuous line, and the way Bach does this so subtly, was really impressive to me.
- 6.38** There's a really nice little passage in the second half of the piece that sounds like this [plays the melody of the passage]. That sounds like a beautiful garland of notes. Nice shape to this line. But what he does is he divides it up into two voices. It's very subtle, but you might not notice, but he does this [plays the passage divided into two voices]. There's no reason that he had to do that, but he couldn't resist. It's like he has to always find these polyphonic textures, even in something very simple like this little two-part texture.
- 6.39** So this line here [plays the line], the E there [plays E] has to be held by the thumb, and it creates this little resonance. Then it splits into two lines. Then the top part holds a C, and it just briefly becomes a little conversation [plays the passage divided into two voices again]. It's like there's this exquisite resonance that emerges out of that that makes the texture so much more interesting. But it's the thing that you're not going to necessarily notice unless you sit down and try to play the piece yourself, and when you add the bass line, you get this wonderful quasi three-voice texture [plays the passage with the added bass line].
- 6.40** We mentioned that in the Goldberg Variations, you have this sequence of variations that are canons. On top of having to produce endless pieces on this one ground bass, he has to turn it into a structure that can support canons at all these different intervals. Some of these are truly technically dizzying.
- 6.41** I'm going to give you an example, which is the canon at the sixth, which is a lovely piece. When I say canon at the sixth, what that means is that in the right hand, I have to play one line with my thumb and index finger and another line with my ring finger and middle finger. It's the same line, but at a distance of a sixth, a fixed interval of a sixth, and staggered in time, staggered by two beats.
- 6.42** The right hand is playing in strict, canonical imitation, and the left hand is playing a free bass line. That's a texture of which Bach was quite fond. It's interesting because what that gives you is a texture that's both very rigorous on one level and also free at the same time. He likes to combine those two things, so a freely elaborated bass line, and then this strict contrapuntal imitation in the right hand.
- 6.43** That brings us back to the first work of Bach that we looked at, which was the Orgelbüchlein, the Little Organ Book. In this case, if I just play you the bottom line at

the right hand, it sounds like this [plays a single canonic voice in the right hand]. If I add the sixth interval, I get this [adds the imitating voice at the sixth]. Then he adds a free bass line, and the bass is like this [plays the bass line]. Then you get this three-part, canonic texture [plays a passage from the canon].

6.44 Now, this idea of the voices chasing and imitating each other at a fixed interval [plays the canon with staggered entries at a fixed interval] creates this wonderful texture of echoes. It's very hard to do because you have to write the canon in such a manner that when it's played along with itself at a fixed interval, it's still harmonically coherent with the bass line. To have done that at every interval on the same bass line nine times, well, that's a hard act to follow, to put it mildly [plays a passage from the canon].

6.45 It's impressive, to say the least, but you don't even necessarily notice it when you're listening to the piece because you're carried along by the rather buoyant rhythm of movement. Every one of these variations has its own distinct character. What happens is, after a while, you forget about the ground bass, and you just notice all of these beautiful, different styles.

6.46 In variation 13, you have this typically elaborate and very ornamented line in the right hand, along with a beautiful figure in the bass that provides the ground for the whole thing. Here we have, again, a three-part texture, and the left hand is going to do this. It's going to play the ground bass [plays the bass line], and so on and so forth, with a second voice that adds a little bit of rhythmic contrast and fills out the harmony [adds the middle voice]. It's a little conversation, but of course, it has to all be done with one hand.

6.47 What's happening here is that the bass plays this stepwise descending line [plays the bass line]. The upper voice plays this figure [plays the upper voice]: a short note, followed by a long note. You add them together [plays the two voices together], and so on. Now we have the right hand, and the right hand plays these wonderful, ornamented figures [plays all the voices together with added ornamented figures].

6.48 The thing that I always am impressed by in pieces like this is this lovely, conversational texture between three lines, each of which has its own particular character. We looked at the smooth descending bass line, the middle voice with its short note preceding a long held note, and then this richly ornamented right-hand line, which is so beautiful. You get absorbed in these textures, and eventually, you stop really paying attention to the bass.

6.49 But that's precisely the thing that a lot of pianists, musicologists, and historians have also criticized about the piece. Glenn Gould, for all that he made his career on this piece, also famously criticized it. He said, "Well, if you don't pay attention to the ground bass and the repetition of that line throughout the different pieces, it's really just a collection of thirty random pieces, each of which is completely different in character from the other." He thought that, as a concept, it didn't quite come off, that the

architecture of it, as brilliant as it is, and for all the musical interests in each individual piece, it can be challenging to sit through all thirty of them. But personally, I love this piece. I can listen to it endlessly, and it's not something that disturbs me at all.

- 6.50** So we've looked at the aria, we've looked at variation 13, we've looked at the canon at the sixth. I think one thing that is pretty clear from this is that they seem to have absolutely nothing to do with the opening aria of the piece. We get very, very, very far from that. The effect of this ingeniousness that he had, of coming up with endless different textures and ideas and contrapuntal techniques and so on, is something that you really have to sit and listen to the piece all the way through to understand. Then you start to, subconsciously, maybe, hear this incredible unity that brings these pieces together, completely transcending the superficial dimension of the opening aria. Not that it's a superficial piece, but in the sense that the successive variations don't resemble it in any way.
- 6.51** The other thing about this ground bass that I'd like to point out, which we looked at a few minutes ago, is that it's a simple binary structure. A binary structure, meaning that it starts, as you would expect, on the tonic in G major, moves towards the dominant, and then Bach has a repeat sign, so you start over again; you play the whole first half of it twice.
- 6.52** Then, once you're at the dominant at the end of the first half of the line, then there's this little moment of circulation and instability, and then it gradually works its way back to G major again. This is a symmetrical structure. In every variation, Bach has these repeat bar lines at exactly the midpoint of every piece.
- 6.53** That's also a question that performers have to think about very carefully, because if you play all the repeats in the Goldberg Variations, the piece is twice as long, and it's already an incredibly long piece. Technically speaking, the score indicates that you are meant to repeat each half of the piece twice. But increasingly, not that many performers actually choose to do it.
- 6.54** There are questions with some of these pieces as to what can an audience take in terms of a tremendous duration of listening. Remember that the audience in Bach's time was most likely a distracted audience, not necessarily listening carefully from beginning to end, but showing up in the middle of a piece and talking to their neighbor and so on, and making noise, and sometimes listening, and sometimes not.
- 6.55** With these pieces, in that era, in 1741, you didn't give public recitals in the way that we do now. You wouldn't all congregate in a concert hall to listen to a pianist or a harpsichordist play a piece like this from beginning to end for 55 minutes or whatever it was. These works, which had a didactic quality to them in the sense that Bach would use it as a demonstration of what's possible within a given form, what's possible within a given genre, were intended for private study and reflection.

- 6.56 You would take a piece like this and try to play it and make that a regular process in your life. You'd have a private communion with these works. Solo keyboard pieces—and this was the case also in the Classical era, well into the Classical era—and chamber pieces were meant to be played by yourself at home for the most part, or possibly at a private occasion. But we didn't have these public concerts in the way we do now.
- 6.57 If you approach it in that way, then it could very well be that Bach never in a million years imagined that you would sit down and listen to all thirty variations from beginning to end, that it was more of a compendium that you would dip into and that had this amazing structure to it, but that you wouldn't necessarily have experienced from beginning to end. It's the same with a lot of the pieces that he wrote during this era of his life.
- 6.58 Now we're getting into the 1740s with these extraordinarily ambitious cycles of pieces, and we're going to continue in the next lectures with, notably, The Musical Offering and The Art of Fugue. We're going to round this one off, and I'm happy to take any questions you may have about the Goldberg Variations.
- 6.59 I'm not a musician, but you mentioned that certain artists can play the same piece basically, very dryly or with rigor. My question is, based on the fact that they're all looking at the same sheet of music and they all have to abide by the rules of what the composer wrote, how does one play the same song with vigor versus dryly? And how, as someone who's not a musician, could I learn to tell the difference?
- 6.60 Composition is a strange art form, because unlike painters who are fixing pigments onto a canvas, there's no intermediary with that. It's a physical object, and the object is also a commodity. It can be bought and sold. You can own a painting, and you can display it in your home, or you can go see it in a museum.
- 6.61 But music isn't exactly a commodity. It's something else. In order for a piece to exist in the world, it can't just be dots on a piece of paper sitting in a drawer. It has to be realized by a performance. It has to be turned into a real thing by somebody playing it. The only way around that, I suppose, is if you make recordings, primarily. Maybe you have a home studio, and you make pieces on your own, and you record everything and put them in a fixed state. It's the same every time it's reproduced. That's the case with recorded songs or any piece of recorded music.
- 6.62 But the tradition, the history of Western art music, concert music, is tied up with the question of notation. The fact that you can take a musical idea, put it down in a fixed form in the form of a score, a written document that can be reproduced by different people in different cities at different times, and you can achieve a permanence of the artwork in that way, but with the proviso that every time it's played, there are going to be circumstances that are going to be different from one performance to the next.
- 6.63 If you factor in not just the differences between performers and their particular state of mind at the time that they have to play the piece, if you also factor in the distance, the

increasing distance from when the piece was actually written, then it gets complicated. Because in Bach's time, there was no piano; the piano hadn't been invented yet. The mechanism of the contemporary concert grand piano was well into the future. Bach was writing for a harpsichord, which is a completely different beast. It sounds very different, has a totally different technique, totally different sonic characteristics.

- 6.64 That's one thing, that the instruments actually physically change over time. So there's a debate in the music world between: should performers only play these pieces on the instruments for which they were originally intended, or is that too limiting? Some performers feel that it's completely wrong to play this piece on anything other than a harpsichord.
- 6.65 Others, like Glenn Gould, who is a pianist, he didn't play the harpsichord, felt that, no, there's something abstract about this music that lends itself to a wide variety of different possible interpretations. There's no reason why you can't play it just as effectively on a piano, and maybe even more effectively, because there's things the piano can do that the harpsichord can't do.
- 6.66 The harpsichord, mechanically, when you press the key, there's a jack that goes up and plucks a string. Because of the nature of that mechanism—it's a plucking mechanism rather than a hammer that strikes a string—you can't affect the sound by changing the velocity of your finger. No matter how hard you press on the key or how quickly you press on it, you obtain the same dynamics; it's the same level of volume.
- 6.67 On a piano, of course, you can play very subtle gradations of intensity, depending on how you strike the key. You can do that with the Goldberg Variations on a piano, but not on a harpsichord. So each instrument has its own particular properties. When you take that into account, then suddenly there's this vast range of possible interpretations, depending on what instrument you play it on.
- 6.68 Then you can try to be as authentic as possible and play the piece as it might have been played by a contemporary harpsichordist in 1741. You can do research on that and look at the instruments and read treatises of instrumental technique from that time, or you can play it whatever way you want. That's a subject of intense debate.
- 6.69 Performers aren't just—there's no objectivity in performance. You're always reflecting the attitude and the sensibility of a performer. There's no way to avoid that. Performers are also artists. What they are is a messenger. They're the ones that are tasked with making the music audible, and they have a personality. Some performers are extremely analytic and precise in the way they play and very focused on texture. Others are always trying to play to the back row in the hall and exaggerating everything and very theatrical in their approach.
- 6.70 Glenn Gould was a supreme virtuoso, and he had technique to spare, and he played the pieces at incredible speed, and others might take a more leisurely approach to them.

There's a broad range of possible interpretations of this piece, and any other piece for that matter.

- 6.71** The other final aspect of it is that tastes change over time. The concert-going public and people who buy recordings also have tastes. They have certain expectations about what a piece should sound like. Record companies and performers, to a degree, respond to that. There was a mania in the '50s and '60s, for example, for making symphony orchestras as large as possible with as many instruments as possible.
- 6.72** A composer like Herbert von Karajan, when he would play a Beethoven symphony or a Mozart symphony would play with an orchestra that was totally not authentic at all, that used contemporary instruments as opposed to period instruments that was much larger than a Mozart orchestra would have been. He created this beautiful, lush sound that was very much of its time, but probably has little to do with how the pieces actually would have sounded at the time.
- 6.73** That's unavoidable. We're always seeing these pieces from a distance. We're hearing them through the filter of our contemporary ears, with what we're accustomed to, and increasingly, we're hearing them electronically through the intermediary of technology. The artwork survives. There's a core of it that doesn't change, but our experience of it is inevitably filtered.
- 6.74** I've got two questions. You can answer the one you prefer or can answer both, as you like. The first one is about, Bach was quite disagreeable, and that's probably in no small part due to all the tragedies he had to go through. I was wondering about how his immense suffering impacted his music. The second question I have is what was Bach's relation to the church, and what is the church's relation to Bach?
- 6.75** Well, Bach clearly suffered. He had twenty children, of whom ten died in infancy, one after the other. That's pretty horrible. He also very suddenly and unexpectedly lost his first wife to an illness. Many of his children, even if they survived into adulthood—he had one child, his name I can't remember because there are so many Bachs; you can't keep them all straight—but one of them was living, I believe, in Berlin and died at the age of 24 from a fever very suddenly. There was just no way to avoid that. That kind of mortality was just omnipresent. The fact that Bach lived as long as he did, reaching the age of 65, was by no means normal. That was an exceptional lifespan. That tragedy was inextricably bound up with life.
- 6.76** From our present day perspective, it sounds pretty awful, but that's what they would have expected at the time. If you look at the lives of all these composers, all these artists, and try to imagine what it would have been like, it can't have been easy for any of them. Beethoven suffered horribly because of his health issues, his deafness, and the fact that he was completely unable to have any stable domestic situation. Beethoven was also an intensely disagreeable person, but to some extent, it's probably just typical of the time.

6.77 And at least, Bach didn't have to live through a massive famine or plague or war, which he narrowly escaped. During his lifetime, it was a period of comparative stability in the region that he lived in. So to what extent did that influence his music? It's hard to say. I can't imagine what it would have been like to lose child after child and still have to get up every day and work. But that's how it was. Thank you.

Lecture 7: The Musical Offering

At a glance — As the course winds down, Andreyev returns to where it began with The Musical Offering, its canons and contrapuntal puzzles.

- 7.1** As we wind up our investigation of the life and work of Bach, in these last two lectures, we're going to return to where we started in the first lecture. I was telling you about my own personal encounters with his music and with the things that got me the most interested, one of which was a work called The Musical Offering, *Ein Musicalisches Opfer*, which is a work that dates from the end of Bach's life, 1747—he was 62 years old. And by this point, he had produced everything you could possibly hope to imagine. He was active in every conceivable musical genre except opera; he had created countless towering works, cycles of complete works, encyclopedic, didactic works. What was there left to do?
- 7.2** Well, by this point in his life, Bach was becoming obsessed with a central idea that characterizes a lot of his production in the later years of his life. The idea was this: Can you have an entire work or an entire cycle of works flow out of a single seed, a little starting point?
- 7.3** We've looked at how previous works of Bach, written in an earlier part of his life, do this, but they do it typically more on the level of the individual piece. For example, in the Two- and Three-part Inventions, we looked at how Bach would take a little short motif and spin an entire piece out of it. I can give you an example of what that means.
- 7.4** We looked at the Two- and Three-part Inventions. The C major Invention starts with this little figure [plays the opening figure from Bach's C major Invention]. The entire piece is built out of that. And if you boil these figures down into what's actually going on melodically, then you can simplify them even further. So that motif there just starts with an ascending diatonic scale [plays the first bar of the opening figure], just a little rising figure, followed by a little dense figure [plays the following bar], a little skipping figure, and then it's rounded off with a cadence [plays the final bar of that figure]. That's really it. I mean, it's hard to think of ideas that are simpler than that [plays the opening figure].
- 7.5** He would spin an entire piece out of that, as we saw, by combining the theme with itself, by creating little sequences, slowing it down and speeding it up and maybe inverting it or flipping it backwards, all the different things you can do with a little motif. So it propagates itself and goes on an interesting little walk.
- 7.6** That was very much characteristic of his work in general. You see it in pretty much everything. You see it in the cantatas, you see it in the instrumental music, you see it in things like the *Orgelbüchlein*. He always had this obsession with development, developing an idea. So, not merely stating an idea, then coming up with a second idea,

and then a third idea, he tries to see all the ramifications of a little, simple starting point.

- 7.7** But in the Inventions, what happens is you go through all these different keys: C major, C minor, D major, D minor, and so on, all the way up the scale. Each piece has a very different character and a different starting point, a different motif. Some of these motifs will signal to the listener, because of their rhythmic character, that this is a dance. Or it might signal that this is a work of pathos and tragedy; this is something very expressive. Or it might be light and frivolous, which some of the Inventions are. But in any case, they were all quite different from each other, and they all had different starting points.
- 7.8** What happens is that as he gets into the end of his life, he starts becoming increasingly obsessed with this idea of, "Can I make an entire cycle of works that's all based around a single idea?" We saw that already with the Goldberg Variations of 1741, where he plans a huge cycle of pieces all around a single ground bass. He starts becoming more and more focused on this idea, more and more methodical.
- 7.9** The composition of The Musical Offering, Ein Musikalisches Opfer, is a very interesting example of this, and the way that it happened is also quite an interesting story. Bach, by this point, has a number of adult children who are pursuing successful musical careers in different parts of the world. Johann Christian Bach, for example, ended up being very successful and popular in London. Carl Philipp Emanuel Bach moved to Potsdam, where he was employed in the court of King Frederick the Great.
- 7.10** So Bach, of course, is proud of his sons and interested in their professional development. And the appointment of Carl Philipp Emanuel Bach in this court was a big deal for him. His son invited him to come and visit and arranged a whole reception for Bach from the king. You have to remember that King Frederick of Prussia was one of the most powerful rulers in Europe at the time. This was a really big deal. And Bach was received like royalty.
- 7.11** When he arrived in Potsdam, the king exclaimed, "Ah, here is the old Bach, gentlemen." He was very excited to introduce everybody to this legendary figure, 62 years old. It turned out that the prince—sorry, it turned out that the king had arranged a surprise for Bach. The surprise was that he had prepared a little theme for him to try out. Remember in the first lecture, I showed you what that theme was, and I explained that it seems to have been devised intentionally so as to cause Bach as much trouble as possible, as a test of what you could actually do.
- 7.12** There's some degree of speculation about this story. I often wonder, "Was Frederick the Great, was he really able to think of such things? Would he have the musical education and expertise to devise this theme, knowing what the difficulties would be for a musician like Bach?" Some people have put forth the theory—Arnold Schönberg, for example, suggested that the theme may in fact have been composed by Carl Philip

Emanuel Bach, cheekily, to try to put his father in the hotspot. It's not clear. I assume that it really is authentically by Frederick the Great.

- 7.13** I'm going to play you that theme again so that you can remember how it goes. Remember that it has a few features that make it particularly tricky for musical elaboration. The first is that it consists of a lot of skips and leaps and dissonant intervals. On top of that, it also contains a full descending chromatic scale covering every semitone between $A\flat$ and B, which is pretty wild. So it sounds like this [plays the royal theme].
- 7.14** On top of all of that, its rhythmic character is odd, too, because there's a whole passage in the middle of that motif where you're not exactly sure where you're at metrically because it's in syncopation; it doesn't quite line up with the meter.
- 7.15** This descent [plays the descending chromatic scale] is inherently harmonically ambiguous because when you have every semitone [plays a sequence of semitones], you don't know what key you're in, and it's very difficult to harmonize things like that. Remember that when you're elaborating on a musical idea, that might involve, if you're working in Bach style, it might involve treating it contrapuntally as a fugue, a canon, or some other imitative procedure. You have to make all of that work harmonically [plays the opening five notes of the royal theme].
- 7.16** Now, that there, that last interval [plays $A\flat$ -B (below)], that's quite a dissonant interval. It's very hard to work with harmonically between the $A\flat$ and the B. That's a diminished seventh [plays $A\flat$ -B (below), then plays the royal theme again]. So, Frederick the Great gives this motif to Bach and says, "I've heard that you're a legendary improviser. Would you like to extemporize on this theme?"—remember, he's in this royal court, in one of the most powerful courts in Europe—"would you like to extemporize on this motif?"
- 7.17** Bach obliges, and because he's Bach, he's able to improvise a three-part fugue right on the spot. What that involves is playing a piece without any preparation, where, first of all, you have to memorize that motif. And then, secondly, you have to begin the piece by playing it, then adding a second voice that also plays it in counterpoint with the first voice, and then adding a third voice. And then they have to all go on and have a three-part conversation.
- 7.18** You're directing three different lines at the same time without any preparation, which is wildly difficult. You can train for that. Of course, there are improvisers that are extraordinarily gifted and can do wild things. But nevertheless, it's quite an impressive feat.
- 7.19** He improvises this three-part fugue, and Frederick the Great is suitably impressed. He says, "Now, can you do a six-part one?" So he sits down and starts to play a six-voice fugue. You're playing three voices in each hand. It's pretty wild. Bach excuses himself and says, "Well, for lack of preparation, I probably need a bit of time to do this." That

seems to be the—even Bach has a limit; improvising a six-part fugue on the spot on this theme is a bit much.

- 7.20** But nevertheless, the visit is a successful one. Bach makes a very good impression. Then Bach returns home to Leipzig and decides, "I've got to prepare a gift, a musical offering to Frederick the Great." So he sets to work, and, characteristically, with extreme industriousness, makes a monumental piece consisting of a three-part fugue on what he calls the royal theme, a six-part fugue—a complete six-part fugue—a set of ten puzzle canons, which we talked about earlier—the puzzle canons are pieces where Bach only gives you a single line, and you have to figure out through what contrapuntal ingenuity can you combine this line with itself in order to create a coherent polyphonic piece.
- 7.21** He also writes a full three-movement trio sonata and a canonic fugue. So it's a substantial work. All of these pieces are based entirely on that motif. He sends it to Potsdam as a gift, and as was the custom in those days, of course, it came with an extremely obsequious bit of text, Bach genuflecting before Frederick in this totally outrageous way. But that's what was done. It was normal for musicians to be extremely obsequious with royalty and to have these very florid, very purple dedications to them.
- 7.22** So this piece is not only a chance for Bach to pay his homage and to express his gratitude for the visit, but there may have also been something opportunistic about it. We don't really know what was his motivation in putting so much effort into this project. Maybe he had hoped for a position in Potsdam. Who knows? Maybe he wanted to increase the reputation of his son, make a good name for his son.
- 7.23** Or was he simply motivated by this interesting challenge? It's like, "Can I take this impossible motif and use it as the single starting point for an entire cycle of pieces?" So once again, he spares no effort. And the piece is famous not only because of the story of its creation, but also because of his ingenuity in finding solutions, contrapuntal solutions to all of these different challenges that he sets for himself. Again, as is typical for Bach, he manages to turn these incredible constraints into pieces of extreme beauty and elegance and refinement.
- 7.24** Now, this is also a compendium of everything you can do in the canon genre. A canon, just to recapitulate—remember, all of these pieces that Bach is working on at this point in his life are a summary of the complete possibilities within that form, whatever it is. If it's variation form, then he does the Goldberg Variations. If he's interested in canons, he's going to do every conceivable type of canon in *The Musical Offering*. If he's interested in fugue, he's not going to just write a couple of fugues; he's going to write *The Art of Fugue*, which is a massive collection of pieces.
- 7.25** This is his treatment of what it's possible to do with canons. There's every canon in this piece. You'll see canons with augmentation, which is where you take a motif and play it at half the speed or twice the speed, diminution and augmentation, and you play it

against itself at multiple speeds at the same time. It's a fractal idea. Or you can flip it backwards, flip it upside down and have the inversion be played together with the prime form, as it's called, or the original form of the motif. All sorts of games like this.

- 7.26** There's one canon in particular that always gets me. It's amazing that he did this. He takes the royal theme, and right at the end, he puts a slight twist on it, where if you're not paying attention, you could easily miss this, but he slightly changes it so that the theme ends a whole step higher than it began. It's a perpetual rising canon—in The Musical Offering, it's called the Canon per Tonos or the canon at the tone.
- 7.27** It keeps rising by a tone every time it loops around on itself, and the music gets higher and higher and higher. It keeps modulating upwards. Bach put a little inscription in the score for Frederick the Great that said, "As the notes increase, so may the fortune of the king." This thing. So these very abstract ideas are also turned into something figurative in that sense.
- 7.28** This is a very interesting stage in Bach's life because he's getting more and more interested in these incredibly sophisticated games. In fact, the original score of the three-part fugue in this collection, The Musical Offering, carries with it an acrostic. An acrostic is a bit of text where you take a word, and rather than spelling it out horizontally as you normally would with one letter after another, you have one letter starting each successive line. The acrostic is where you take each letter of a word and use it as the beginning of another word, and so you can create a little text out of it.
- 7.29** In the case of The Musical Offering, rather than referring to the three-part and six-part fugues as fugues, he used another term—ricercar, R-I-C-E-R-C-A-R—which is a somewhat difficult term to define precisely because it has different meanings in different contexts and in different countries, but he called this piece Ricercar a 3, Three-Voice Ricercar, which incidentally means research. It implies that the piece has a—what would you say?—a quasi-scientific dimension to it, where you're trying to push the limits of something, test an idea in a sense.
- 7.30** He takes the word ricercar, R-I-C-E-R-C-A-R, enters it in an acrostic in Latin, which is Regis Iussu Cantio Et Reliqua Canonica Arte Resoluta, which means, "According to the king's order, the tune and the remainder are resolved with canonic art." In other words, he's received this request from Potsdam, and he's fulfilling it in resolving all the potential of this line through his contrapuntal wizardry.
- 7.31** Let me show you what happens in the Canon per Tonos. Here's the motif, which you remember [plays the original motif]. What he does is he slightly modifies the end [plays the altered ending of the motif] so that now the theme ends on D rather than C. And he also modifies the beginning a little bit. It's still definitely the same motif, but he tends to find different ways of presenting these motifs by adding passing tones like [plays the motif with added passing tones]; he adds a little ascending scale so that instead of the

original motif [plays the original motif], he adds all these intervening tones [plays the elaborated version of the motif]. It's an elaborated form, if you like, of the opening.

- 7.32** But the really important thing here is how he throws in that little change at the end [plays the altered ending of the motif] so that it ends higher than it begins. It's so sneaky that you don't even notice it while it's happening. But the way that this piece is played is you just keep repeating it until it gets higher and higher and higher. In principle, it could just go on forever until you run out of notes on whatever instrument you're playing it on.
- 7.33** It's a very clever bit of ingenuity. It's like there's this thing in acoustics called the Barber Pole Effect, where you can have those poles that they have in barber shops, where it keeps rising and rising and rising, and then it starts up from the bottom again, perpetual rising motion. That's what this is.
- 7.34** Let's start by listening to the Canon per Tonos from the Musicalisches Opfer [recording plays].
- 7.35** So what do you get if you pursue this idea to its limits? You get a form of infinite recursion, where the idea keeps going and going and reproducing itself. In this case, this canon that endlessly rises forever, as long as you want to keep the piece going. It's amazing. The problem, I suppose, is that King Frederick was not very interested in this style. His own personal tastes were a little bit different.
- 7.36** By this point, the 1740s, this old style, this contrapuntal style, was pretty much thoroughly out of fashion and was beginning to be displaced by something quite different, which was called the galant style. The galant style was characterized not so much by these contrapuntal textures and by the use of continuo, the use of a harpsichord or a bass instrument playing together and realizing a figured bass; that was starting to disappear.
- 7.37** Instead of that, you had these much more transparent textures characterized by a simple, clear melody in the right hand or in the melodic instrument, whatever it was, and then little accompanying figures in the left hand. This was a simplification in some respects relative to the older style in which Bach spent most of his career writing.
- 7.38** It was called the galant style because it was pleasing, and it was easier to listen to and could be extraordinarily beautiful. That necessitated a sharper division, if you like, between a principal voice and something subsidiary that supports the principal voice. We're no longer thinking in terms of autonomous lines.
- 7.39** Remember, one of the principles of counterpoint is that the lines are all of equal significance. It doesn't matter in a sense, if you're playing one of these pieces, if it's a bass line or a soprano line or an alto line, they all have equal importance in this texture, and they all equally contribute to its elaboration. It's not so in the galant style.

- 7.40 Bach's own son, Carl Philipp Emanuel Bach, historically is a very interesting figure because he existed in this transitional period between the Baroque and the galant style, which would ultimately lead to the High Classical style of Mozart and Haydn. So he's a transitional figure. Carl Philipp Emanuel's music was much more in keeping with the tastes of the king.
- 7.41 I thought it would be interesting to play for you a piece by Carl Philipp Emanuel Bach, a short movement from one of his keyboard sonatas, that illustrates this gap. This is a piece, Sonata in G major, that was written at around the same time as The Musical Offering. It gives you a chance to see the difference between what Bach was doing at the end of his life and what his very successful son was writing at the same time. Also, it gives you a glimpse of the musical future after Bach, of the way things would be evolving in Europe in the mid-18th century.
- 7.42 Bach was immensely respected. He was respected primarily in many circles as an organist and as an improviser. But he was not necessarily a composer whose works were highly sought after by people like King Frederick at that stage. They recognized his artistry and his supreme ability. But tastes changed, fashions changed, and he was more interested in this other style.
- 7.43 Let me play an excerpt from a Keyboard Sonata in G by Carl Philipp Emanuel Bach [recording plays].
- 7.44 The difference in style is pretty striking. Bach, at this point in his life, is writing this very severe music in a way. This is something completely different. It feels almost improvisatory. It's very free, all kinds of different rhythmic figurations. It's very gracious, very graceful, very melodic. That was really the future at that point. What Bach represented was, well, at that point, it really belonged to the past for all of his ingeniousness with these contrapuntal elaborations.
- 7.45 I want to show you also what these three-part pieces sounded like in The Musical Offering, to go back to this piece again after that parenthesis about the galant style. So I mentioned that Bach extemporized a three-part fugue when he was in Potsdam—I'm not going to go into too much detail in terms of explaining what a fugue is because that's going to be reserved for the last portion of this course, where we talk about The Art of Fugue.
- 7.46 But essentially, a fugue is a polyphonic composition involving different voices that imitate or chase each other, if you like. This is—we are told—a written-down version of the improvisation that he produced in Potsdam. So Bach comes up with something on the spot, later writes it down somehow—I don't know how he managed to remember it all, but apparently that's what happened—and produces this three-part fugue.
- 7.47 It's very interesting to compare the three-part fugue to the six-part one because the six-part one is the one that he had to actually work out very laboriously at home when

he got back to Leipzig. Indeed, the three-part fugue, which is a lovely piece, but it's not as perfect, it's not as refined as the six-voice one.

- 7.48** I'm just going to play you the beginning of it so you can see what's involved in taking a motif like this, playing it against itself, and trying to find contrapuntal combinations that work [plays the opening of Bach's three-part fugue]. That's pretty wild, that counterpoint there because he's got all of these huge leaps like this [plays a passage from the fugue featuring wide leaps].
- 7.49** There, you can hear the three-part texture starting to come in. Each voice begins with the same motif, first in the top voice, then in the middle voice, and finally in the bass [plays the bass line], and so on. You have to keep all of those voices going. It gets quite complex as a texture very quickly. That he was able to improvise on the spot, apparently. The six-part *ricercar*, the six-part fugue that brings *The Musical Offering* to a close, is a piece that has fascinated musicians for centuries now. It is, in a sense, the ultimate crowning glory of his achievements as a contrapuntalist.
- 7.50** I'm going to play you the six-part *ricercar* from *The Musical Offering* in a harpsichord version, and you can see what it sounds like [recording plays].
- 7.51** Thus concludes Bach's monumental investigation into the possibilities of both canon and fugue, based on a single theme, the entire piece. That piece has been arranged many times by subsequent composers who have produced versions for string quartet, for orchestra, for all kinds of different arrangements. If you're interested in it, you can go online.
- 7.52** It's actually quite interesting to hear different versions because you can hear the lines a little bit more clearly if they're sustained by string instruments or wind instruments. The thing with the harpsichord is that it has a very rapid decay time, so there's not much sustained to the sound, which can make it harder to distinguish the different voices.
- 7.53** You mentioned a few times that this musical motif is quite awkward and difficult, and tricky one. Do you think Bach would have been able to do the same thing with any motif, or did the awkwardness of the musical motif actually help him, like he had more material to work with? Or knowing Bach, maybe he would have been able to do the same thing with anything, basically?
- 7.54** I think it must have stimulated his musical imagination, definitely, because the pieces that he produced in this collection are remarkably ingenious and don't really sound like anything else he produced. The constraint and the difficulty of it, I think, were things that challenged him creatively and forced him to come up with interesting solutions.
- 7.55** Bach could certainly take motifs that are very, very difficult to work with and turn them into something completely fluid. He did that at the end of his life, too—we'll talk about this a little bit more in the next lecture when I talk about *The Art of Fugue*.

- 7.56** But one of the things you can do in music is, because the names of the notes are associated with letters of the alphabet, you can hide messages within a musical piece: C, D, E, F, G, A, B. Or in German notation: C, D, E, F, G, A, H—B in German is B $\flat$ and H is B natural. We won't get into the reasons for that right now. But you can have little ciphers, little encoded messages in a piece by using the letters of the alphabet.
- 7.57** What Bach realized was that his own name spells out a series of four tones: B—so B $\flat$ in the German system—A, C, and H, which is B natural. That's Bach's musical signature, if you like [plays B, A, C, H]. He uses that as a motif in the final fugue of The Art of Fugue.
- 7.58** Now, that's a motif that is ambiguous in the extreme because it consists of four adjacent semitones [plays A, B $\flat$, B, C]. It doesn't really imply a tonal center at all. It's not easy to work with materials like that. So even just a little chromatic fragment like that, he's able to turn it into a workable motif, and it's pretty amazing.
- 7.59** So when you hear things like that, you hear The Musical Offering, and also that chorale prelude that we heard, Durch Adams Fall ist ganz verderbt, where he had that crazy chromatic bass line with all those leaps. Remember, it sounds like it's in no key at all, and he managed to write a completely plausible and coherent piece, a very expressive one, but working with a line that, if you play it on its own, seems to make no musical sense whatsoever. So he was extraordinarily clever, and it's tempting to imagine that he could have turned anything into a musical composition.

Lecture 8: The Art of Fugue

At a glance — The final lecture treats The Art of Fugue as the summit of Bach's contrapuntal art, and reflects on the selection of works across the course.

- 8.1** As we round off this final lecture in the course on Bach, I want to just point out that the selection of pieces that I've made here was chosen in part to illustrate the different periods of Bach's career as his priorities shifted, moving around from position to position, from city to city. But because of the constraints of trying to fit this all into eight hours, of course, there are many, many major works that I haven't had time to get into in detail.
- 8.2** Many of the pieces that we haven't touched upon here are secular instrumental pieces that are enormously appealing and great fun to listen to. Things like the Brandenburg Concertos, for example, which, again, are such beautiful pieces to listen to if you get a chance to hear them. There's all the great instrumental works of his years at Weimar and the cello suites. Those are pieces that you may already have heard. In fact, they're very popular. But I simply can't get into detail about all of them; there are so many.
- 8.3** But we do have to absolutely talk about The Art of Fugue, which is one of the last pieces Bach ever composed. This is a piece that is quite enigmatic in many respects, and he worked on it throughout the last decade of his life. We don't know precisely when he started work on this piece, but most likely in the early 1740s. Bach made it to 1750, and he died at the age of 65. So this was a piece that was in gestation for quite a lengthy period.
- 8.4** But we do know that the central idea of this work, *Kunst der Fuge*, The Art of Fugue, was to do what he did for The Musical Offering, effectively, which is to take a particular process or genre or form and show everything that he was, by that point, able to do with his musical knowledge, his command of the science of counterpoint. In this case, the subject of his inquiry was the fugue.
- 8.5** At this point, we really have to define what a fugue is. This is something that Bach was preoccupied with throughout his entire career. He wrote hundreds of fugues. What exactly is a fugue? A fugue is a contrapuntal composition that typically starts with an expository section in which you begin by playing a fugue subject.
- 8.6** A subject is a short musical motif that is played successively by however many voices there are in the piece. If it's a four-voice fugue, then the subject will be heard four times, typically in tonic-dominant, tonic-dominant alternation. So you'll hear the initial version of the subject on the tonic. Then the second voice will come in while the first voice is still playing, and will play the same thing but transposed up a fifth. Then the third voice will come in and play it again on the tonic. Then the fourth voice will enter, completing the cycle and starting on the dominant. You alternate tonic-dominant.

Then, gradually, you add all the voices together. You start with one, and then two, and then three, and then four.

- 8.7** The term fugue means flight. One of the things that it tends to sound like when you're listening to a fugue is the voices having a conversation with each other, and at times getting closer and closer together and creating an impression as though they're almost chasing each other.
- 8.8** This was a medium that particularly suited Bach's way of thinking. To say that counterpoint was important to Bach would be as banal as saying that water is important to fish. It was the entire basis of his technique. It was the technical basis, not only of his music, but of art music in general, when he began his career. In short, counterpoint was really everything. It was the entire foundation of his approach to composition. Fugue allows ideas to be elaborated contrapuntally in a particularly clear way.
- 8.9** One of the things that fugue allows you to do also is to show all of the latent possibilities that exist within a single idea. When Bach embarked upon the project of writing *The Art of Fugue* in the last years of his life, the idea was to present a summary of all the different possible contrapuntal combinations that you could possibly think of, starting from a simple subject.
- 8.10** I'm going to start by playing you the subject of *The Art of Fugue* [plays the first four bars of the opening section of Bach's first fugue], a very short little four-bar phrase. Now, this subject has a few characteristics that are worth pointing out right at the outset. The first is that it starts by outlining a D minor triad [plays D, A, F, D]. If you just take those three tones and play them together, you get a D minor chord [plays a D minor chord]. What that means is it immediately establishes the tonality or the key of the piece straight off the bat.
- 8.11** Then it moves downwards [plays the descending continuation of the opening]. The other interesting characteristic of this little motif is that it starts off with a rhythm in minims or half notes, and it seems to gradually speed up as it goes along. Then it moves towards quarter notes or crotchets and finishes with quavers or eighth notes. If you listen to the rhythmic profile of this subject, notice how it starts with longer note values, and they progressively get faster [plays the first four bars of the opening section].
- 8.12** Now, another attribute of this subject is that it fits into a very small space. The total range that it covers spans only a sixth, from C# to A [plays C#–A, followed by a few figures in that interval range]. The whole thing fits into a closet, basically. It's a very compact little subject. He chose it for a few reasons. One of them is that because of the variety of different rhythmic values in the subject, you can combine it with itself at different speeds in different ways to create a beautiful texture of interlocking rhythms. You can have rhythms in longer note values, rhythms in quarter notes, rhythms in

eighth notes, and dovetail them, have them all together. Already when he invents his subject, he's thinking in terms of: How can I combine it with itself to create a rich texture? Of course, one of the components of that is rhythm.

- 8.13** I'm going to start by playing you the expository section or the opening of this fugue, so you can get a sense of what this—I hesitate to use the word genre. Fugue is not exactly—it's neither a genre nor a form, really, but it's more like a way of thinking or a process whereby you create a conversational four-part texture all based around one idea [plays the opening section].
- 8.14** So hopefully you can hear that progressively, he keeps adding more voices. He starts with the alto, in this case, the middle voice. Then there's a soprano voice that enters on top of it [plays the alto line, then adds the soprano line]. Then he adds the bass [plays the alto, soprano, and bass together]. Then finally, once he's got the three voices in play, the fourth voice enters the texture [plays all the lines together], and so it continues.
- 8.15** The thing that really just astounds me in all of these pieces is—and this is in keeping with what we saw with the chorales and the Orgelbüchlein and all of these different pieces—you can isolate any of these lines, and it's a beautiful, complete melody in and of itself, almost without needing the accompaniment of the others.
- 8.16** If we strip away the soprano line when it enters and just focus on the evolution of the alto, then you can see that it's a completely self-sufficient line [plays the alto line], and so on. Now, let's try it with the two voices together once again [plays all the lines together].
- 8.17** Now, what's happening here is once all four voices are in play together, you'll have a little series of episodes. Episodes are short little portions of the piece that might be characterized by a sequence. A sequence is taking a short motif and having it rise or descend by step or according to some other harmonic pattern. You're going to notice the same little motif played multiple times in slightly different ways.
- 8.18** There's a nice example of this here [plays a sequential passage from the fugue]. If you were to isolate just the top line in the right hand, you would hear this [plays the soprano line]. You notice that it's just a very short little idea, but it repeats itself slightly higher each time. And likewise, if you isolate the bass line [plays the bass line], there as well, we have a little sequence, in this case, a chromatic sequence. I love this little line. It's easy for it to get buried in the texture, but it's actually quite a wild little piece of writing because it contains all of these different chromatic semitones [plays a sequence of semitones from the fugue].
- 8.19** What's happening is that Bach animates these fugues by having very progressively changing registers. He'll start a voice maybe at the top of its range, and over many bars, the line will gradually descend. There's constantly this sense of movement in every

voice, where they're moving towards the bass or moving up towards the upper end of its range. This gives the piece a wonderful sense of fluidity.

- 8.20** In the case of this bass line, you can trace that descent down over about eight bars. I'm going to isolate this bass line and notice how it evolves over time [plays the bass line]. There's a rising figure—it winds upwards like the proverbial barbershop pole. Once it reaches the apex, which is this high D [plays B, C#, D, D], then it tumbles downwards two octaves. It goes all the way down to here [plays D, F, E, D, A].
- 8.21** If you listen to the texture of the passage as a whole, you can see that it's orienting your perceptions with this inexorable move from one register to another. It gives it a wonderful sense of movement [plays the passage, tracing a gradual rise followed by a long descent through the registers]. You can see this constantly moving line that wanders all over the piece.
- 8.22** I mentioned that there's all kinds of ways that you can take a little motif like that and present it musically. You can combine it with itself, you can play it at different speeds, and you can also do something called inversion. Inversion is a contrapuntal process that involves taking the line and essentially making a mirror image of it. So if you have an interval that's a rising interval, like a perfect fifth, that goes up from D to A [plays D–A], then you invert it so that the line goes in the opposite direction [D–A (below)], and what you end up with is a mirror image of the motif.
- 8.23** In *The Art of Fugue*, Bach does this in several of the fugues. He takes the motif, makes a mirror image of it, and then uses that as a derived fugue subject and builds a piece around it [plays the opening section of the third fugue]. That's the beginning of the third fugue in *The Art of Fugue*.
- 8.24** You can hear that it's the same subject. You recognize the same shape and the same rhythms, but it's flipped onto itself [plays the first few bars of the opening section of the third fugue], and so on. Hear how it combines itself when he adds the second voice [plays the entry of the second voice against the inverted subject].
- 8.25** It's a summary of this obsession, this drive towards unity, unification, of seeing all the consequences you can pull out of a single idea. In fugue writing, you can flip the image around an axis and obtain an inversion of a fugue subject. You can play it at half speed, you can play it at double speed, and you can also retrograde it. Retrogradation is taking a subject and playing it backwards, as we saw in *The Musical Offering* with the famous Crab Canon.
- 8.26** There's no retrogradation in this piece. That's also something that is an interesting clue into Bach's way of thinking at this point in his life, because I keep saying that these pieces have an encyclopedic dimension to them, where Bach is trying everything you can possibly do with these subjects. But he's still an artist, and so it still has to work musically. It has to be musically coherent. He's not going to do things in these pieces that don't make musical sense.

- 8.27** In The Art of Fugue, you will see inversions and augmentations and diminutions and various permutations of this motif throughout the piece, but you're not going to see a retrogradation. The simple reason is that retrogrades can be quite difficult to work with contrapuntally. There's no necessary reason why a motif played backwards is going to result in something musically valid or interesting. In this case, if you were to play the subject backwards, it would be actually excessively awkward and not particularly musically interesting, so he avoids doing that.
- 8.28** Nevertheless, we've looked at all the different ways that Bach manages to exhaust his material and to push his artistry and his sophistication into ever more recondite corners. In The Art of Fugue, what he does is no less astonishing. His ambition seems to have been to create a cycle of fourteen pieces—we assume it was fourteen—that provide a very comprehensive overview of the possibilities of fugal treatment, but he was unable to finish the piece.
- 8.29** The final fugue in the collection is number 14, Contrapunctus XIV, as he calls it, and it's an incomplete piece. It breaks off on the final page. It's not clear if the piece was left unfinished because he was no longer able to continue because of his health, or if it was in fact completed or planned out and we're simply missing a page of the score, or if he ran into some difficulty with it, or if he wasn't quite sure how to finish it. I think that last example is probably the least likely. But nevertheless, The Art of Fugue, his monumental exploration of fugue possibility, is in fact an incomplete work.
- 8.30** There's approximately, I think it's seventeen bars that are thought to be missing from the final fugue. When you're listening to it, some versions, some recordings will just abruptly break off. The voices stop, and the piece doesn't come to a conclusion of any sort. Some musicologists have attempted to complete the final fugue by imagining what he might have done had he pushed it all the way to the end.
- 8.31** But the intention seems to have been to write a fugue with multiple different subjects that are combined with each other in different ways that, unfortunately, we will never know exactly what he would have done right at the end. But it was towards the end of that piece that he started to introduce his own name into the fabric of the piece: B-A-C-H. And Carl Philipp Emanuel Bach, when he was preparing the manuscript posthumously for publication, wrote a note in that first edition of the piece that it was at this point in the final fugue that Bach started to integrate his own name into the piece as a subject, and that was the point at which he died. That may be an elaboration. We'll never know for sure. The circumstances of Bach's death are tragic, but they're interesting to get into.
- 8.32** Bach may well have suffered from diabetes, and towards the end of his life, he started to have vision problems that were proving to be increasingly troublesome. So he was being prevented from working by the fact that it was now getting to be extremely physically uncomfortable, and he simply couldn't see well enough to keep going.

- 8.33** As fate would have it, there was an English doctor who was passing through Leipzig, who was famous for his radical and experimental surgeries for eye problems. The decision was made to have Bach undergo this surgery on the slim hope that maybe this would cure him. I don't even really want to get into the gruesome details of what this operation entailed because it was pretty horrible. Of course, it would have been done without an anaesthetic at the time. But suffice it to say, this operation was not successful, and Bach's health rapidly declined afterwards, and that's what killed him. It was a botched eye operation, actually.
- 8.34** He dies at the age of 65 with his magnum opus not fully completed. But nevertheless, even in its incomplete form—and really not very much of it is missing; out of fourteen fugues, there's a few bars of the final one that we don't have. In terms of what's there, it's a remarkably rich piece, inexhaustible as it were.
- 8.35** There's a further aspect of The Art of Fugue that has contributed to it being considered a very enigmatic piece—something I mentioned at the outset. One of the central enigmas of the piece is that we don't exactly know what his intentions were for it because the score, as he prepared it, was written in what's called open score. Rather than writing it on the two staves of a grand staff for what would be conventional keyboard notation, Bach wrote The Art of Fugue out on four staves using the old soprano clefs, these different clefs that were used for a soprano, alto, tenor, and bass. So he writes it out as a pure piece of abstract music without any clues as to what the instrumentation was intended to be.
- 8.36** That in and of itself is not particularly exceptional because in the Baroque, you would play pieces on whatever instruments were available, and lots of composers wrote pieces that were intended to be transcribed or played on different instruments. Bach's contemporary, Telemann, for example, wrote fantasias that can be played on recorder, bassoon, flute, oboe, or violin. It doesn't really matter. It's the same basic piece either way. In and of itself, it's not that unusual, but Bach did take instrumentation seriously.
- 8.37** In works like the St. Matthew Passion, there's no denying that the color and the orchestration and the use of exotic instruments like the oboe d'amore and all these different voice types, the flutes, and so on contribute very much to the meaning and the emotional resonance of the work. In the case of The Art of Fugue, we simply don't know how it was intended to be performed or even if he ever intended it to be performed; if perhaps this piece was a pure piece of abstract music, somehow.
- 8.38** Many people have said that The Art of Fugue is a piece that, if you can read music, you can just pick it up and read it to yourself. Maybe that's ultimately the point. Maybe that's the purest way to experience this music is to try to read the line silently yourself and see this abstract, idealized version of music as it appears before your eyes. Maybe any attempt to play it on an instrument can only take it away from that idealized version, that idealized platonic existence of the piece. But whatever it may be, the score as it exists is a total abstraction.

- 8.39** Now, there are theories that it fits better on an organ, or it fits better on a harpsichord, or it works within the limited range of this instrument or that instrument, and so he must have had an instrument in mind, but we don't exactly know. The fascinating thing about it is that this piece has this extraordinary level of abstraction, contemplation. It's a pure play of pattern and derivation of a single idea in all these multiple different forms. Whether it's done on a string quartet or four human voices or four saxophones or a piano or a harpsichord or a synthesizer, the message is the same. In a sense, this is the apotheosis of the possibilities of just pure abstraction in music, but in this sublime form.
- 8.40** Before we round up this series of lectures, and before we leave off on the subject of Bach, I'm going to play you organ performances of two fugues from The Art of Fugue. Actually, I say two fugues; it's really only one—I'll explain what I mean. As Bach is going along working through his cycle of pieces, he decides to write a mirror fugue. The mirror fugue is, again, one of these things where, on paper, it sounds impossible. But he writes a fugue where if you take each voice and permute its position—so that the soprano becomes the bass voice and the alto becomes the tenor voice, and so on—and you flip the lines around, you make a mirror image of each of them, then you get a second fugue that works equally well. It's amazing.
- 8.41** That is very, very technically hard to do. It's almost ridiculous. And he does it, of course, and so you have this fugue that exists in two versions: the original version and the mirror version, where everything is basically upside down. I want to play you the famous mirror fugue from The Art of Fugue [recording plays].
- 8.42** Now we're going to hear the mirror image of that. Just for the sake of interest, this performance is by a German organist named Helmut Walcha, W-A-L-C-H-A. He was blind from birth. Walcha recorded the complete organ works of Bach, learning each piece one voice at a time, one note at a time, and memorizing them. Astounding. So here's the inverted image of this fugue [recording plays].
- 8.43** The Art of Fugue was published in 1751, the year after Bach's death, and sold fewer than 20 copies. It was not a runaway success. We have time to take a few questions before we round off this course.
- 8.44** Could there be another composer like him?
- 8.45** Probably not. There hasn't been so far. I've thought about this question, too. I think that what produced Bach was a whole culture, really. There was a whole ferment. There was the extended Bach family—the fact that every member of his family was a professional musician—the fact that he existed in a time that was briefly calm, relatively stable, that he was able to live out his life within a very limited geographic region with a minimum of turbulence, it has to be said.
- 8.46** Obviously, he had his personal and professional troubles, but there's nothing near as bad as it might have been. That combined with his genius, his relentless application, his

tremendous this conscientiousness as a human being and as an artist, and having the conditions and the work situation also that pushed him to work as hard as he did, you'd have to replicate those conditions, and I don't think you can.

- 8.47** I think for artists to thrive, someone needs to need what they're doing. There needs to be a demand for it on some level. I think that art forms shrivel up if there isn't enough interest, if there isn't a function that it fits into, at least on some level. Having had to write hundreds of cantatas, all of these fugues, and all these pedagogical projects, he had a mission that was a personal artistic mission, but that also dovetailed with a period where there was intense demand for that, also in an era where the depths of the art form had not yet been completely plumbed. There was a lot of exploration to be done.
- 8.48** You see that there's a life cycle with different genres and forms and artistic disciplines. In order to reach a height like that, I think a lot of things have to come together. There have been other equally profound artists, of course. There are many great figures in the history of music, each of which responded to the particularities of the time in which they existed, and that will continue to happen. We're going to keep getting artists that reflect the times that we live in.
- 8.49** There's a German composer who died a few years ago who was immensely productive, called Rehm, R-E-H-M—I think I mentioned his name already earlier in the course. The reason I mentioned him again is that he was one of these people who had that capacity to just sit and work and write nonstop and produce a huge body of works. But increasingly, I wonder who's going to do that because it's a difficult way to live, especially if nobody's asking for it. I'm not at all pessimistic about the future of the art form, but I think it's going to take very different forms moving forward.
- 8.50** I was wondering if you have a sense or if there's a sense based on the analysis of Bach's work of what makes for a good or a promising initial seed for a fugue piece?
- 8.51** Amenability to transformation—that's the first thing. Is the starting point an idea that's flexible enough and simple enough so that you can bend it and stretch it and pull it apart and combine it with itself in hundreds of different ways. That's certainly very important.
- 8.52** But what's even more important than that is the capacity of the artist to see potential within material, just as a great sculptor can look at a block of marble and imagine a statue of David within it. It's important, I think—and this is a difficult thing for younger composers to get a handle on—the important thing is not the material itself. It's trivial to find material. The important thing is to develop a way of handling the material so that you can see the potential within it and realize that potential.
- 8.53** Bach is one of the more extreme examples of that, I think. He had an unbelievable combinatorial mind. He could run through all of these possibilities at rapid speed and imagine what you could do. That being said, it doesn't mean that he could write things

like The Art of Fugue completely without any preparation or just to casually dash them off. These were pieces that he worked very hard on over a period of about possibly 10 years or more.

- 8.54** Some of these pieces are so complex in their contrapuntal elaboration that there's no way that he could possibly have written them without extensive preparation, trying out different subjects and seeing what the combinations would look like, especially if you're going to do something like a mirror canon. That's really pushing it to the absolute limits.
- 8.55** But yes, amenability to transformation is an important one. Another example of that is the music of Beethoven. If you look at Beethoven's themes in his symphonies and sonatas and string quartets, the themes are nothing; they're so trivial. They're completely uninteresting. It'll just be a little rising arpeggio. The opening theme for the C minor symphony, which everybody knows [plays the opening motif of Beethoven's Fifth Symphony], is not a melody, whatever that is; it's a motif, maybe. It's four notes. The F minor piano sonata [plays the opening arpeggiated figure from Beethoven's F minor piano sonata]. That's his subject. It's just a rising arpeggio. The F major string quartet, Op. 18, No. 1 [plays the opening motif of Beethoven's F major string quartet].
- 8.56** The interest is not in the material. The material is not interesting. It's the architectonic dimension of the piece. It's the way the material is spun out, the way it develops, the way it changes, the way it's extended over time. That's the hardest thing to do in music. That's where all the skill lies. You see these artists and their capacity to take seemingly anything and use that as the pretext to make a great piece.
- 8.57** In my Introduction to Composition class, we started out by looking at two paintings of apples: a painting by René Magritte, the Belgian surrealist painter, and a painting by Philip Guston, the American abstract expressionist painter. They're both of apples, but they're completely different paintings. The point being that nobody cares about the stupid apple; that's not the point. The point is: How do you handle the material? How do you handle the subject? Can you do that distinctively?
- 8.58** Again, you can go into any museum in Europe and see endless paintings of Saint Sebastian being pierced by arrows, but no two of them are the same. And some of them are way more interesting than others because the painter had the capacity to invent something striking and unforgettable using the subject as a pretext. It matters, and it doesn't matter. It matters in the sense that there are a lot of consequences downstream of the original subject, whatever it is, whatever you start with, that will change what you end up doing. But a great composer, a great artist can start with just about anything.
- 8.59** We've listened to several different recordings over this course. You'd highlight it in particular Glenn Gould's Goldberg Variations. Are there any other recordings you'd

particularly highlight out is worth listening to for any of the pieces we've listened to over this course or any others.

8.60 Yeah, absolutely. There's a recording of the St. Matthew Passion by a conductor named Herreweghe. He's a Belgian conductor. H-E-R-R-E-W-E-G-H-E. I think it's Philippe. Philippe Herreweghe. His recordings of Bach are, especially the vocal works and the orchestral works, absolutely superb. All of them. He plays on period instruments. In fact, I think the extracts that you heard from the St. Matthew Passion were his recording. They're absolutely top-notch.

8.61 As far as keyboard works go, I highly recommend listening to Helmut Walcha, W-A-L-C-H-A, and his versions of the organ works, which are simply extraordinary. I would probably start with those.

8.62 If you want to listen to the great keyboard works, The Well-Tempered Clavier, the Goldberg Variations, the French suites, the Italian suites, and so on, I do recommend Glenn Gould. He's not to everybody's taste. He was considered a very eccentric performer at the time, and he was also heavily criticized for playing on a piano as opposed to a harpsichord, but if you can get past all of that, they are recordings of extraordinary dexterity and imagination and wit and virtuosity, and they're tremendous fun to listen to. Those are a few recommendations, but the Bach discography is extremely vast. There's no shortage of great versions of all of these works.